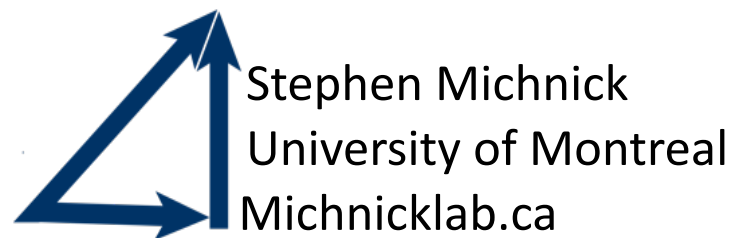


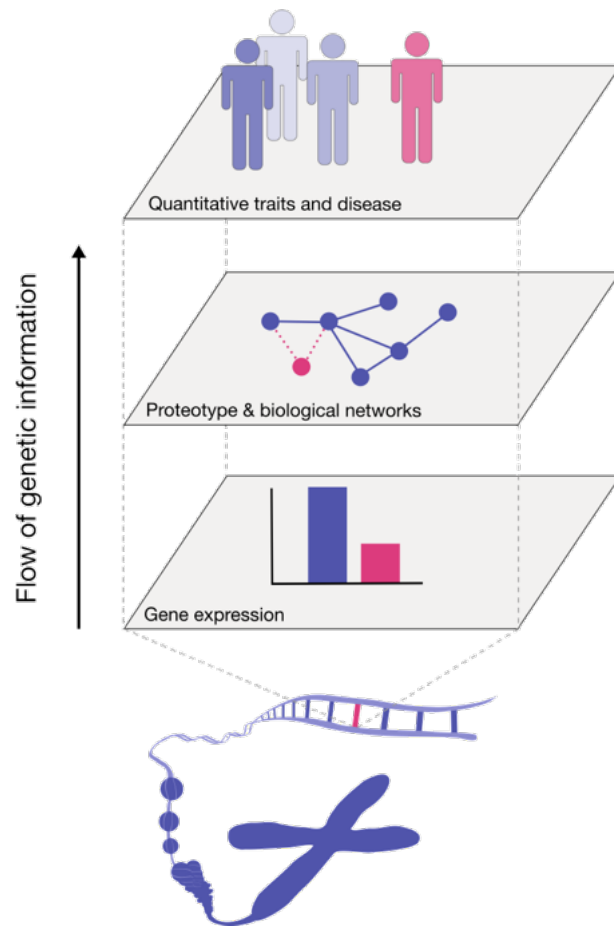
The genetic landscapes of an *in vivo* protein interactome

Single-Molecule & Spatial Sequencing: Methods & Applications (Spring 2025)



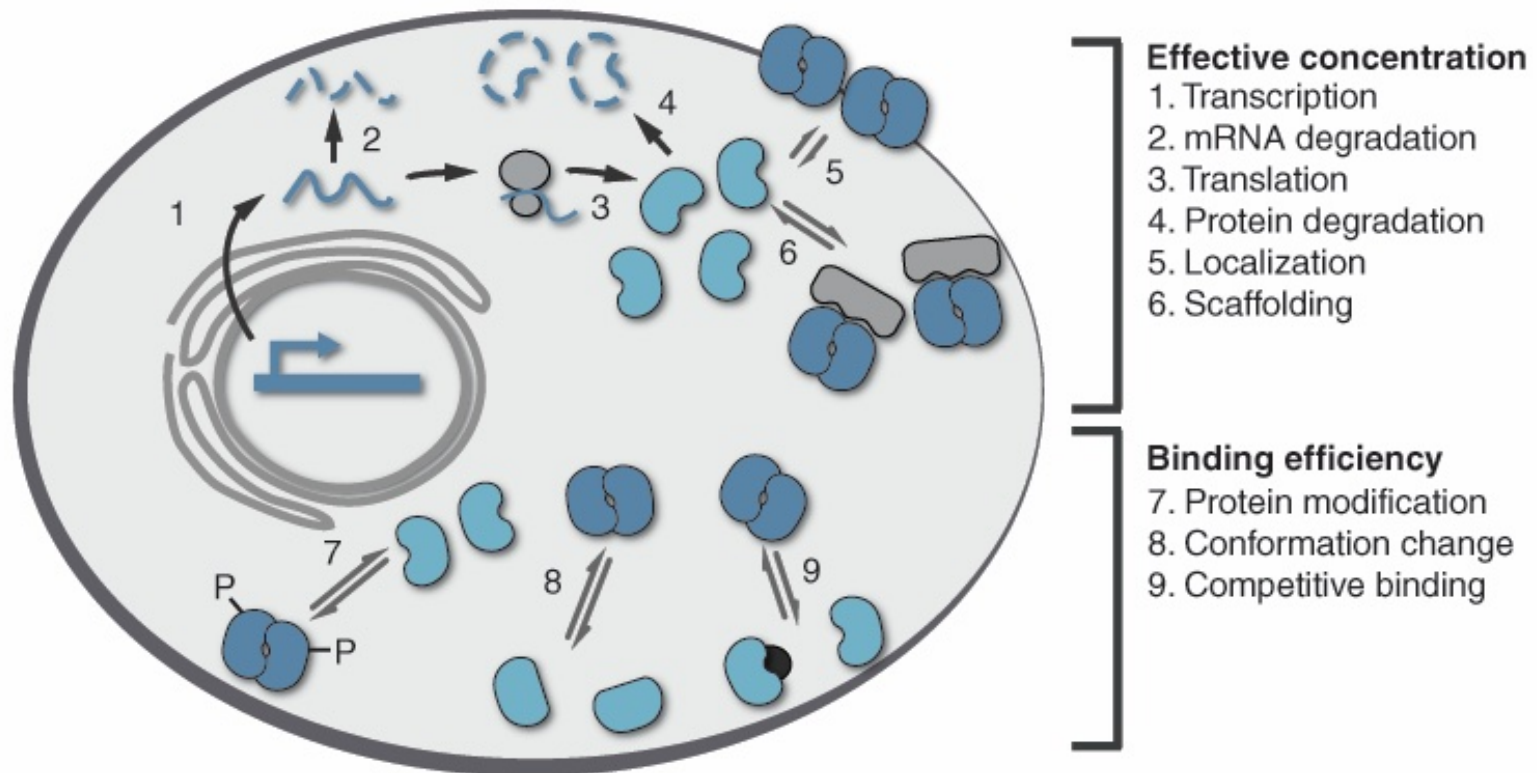


Biochemical networks link genotype and environment to mechanisms of variation

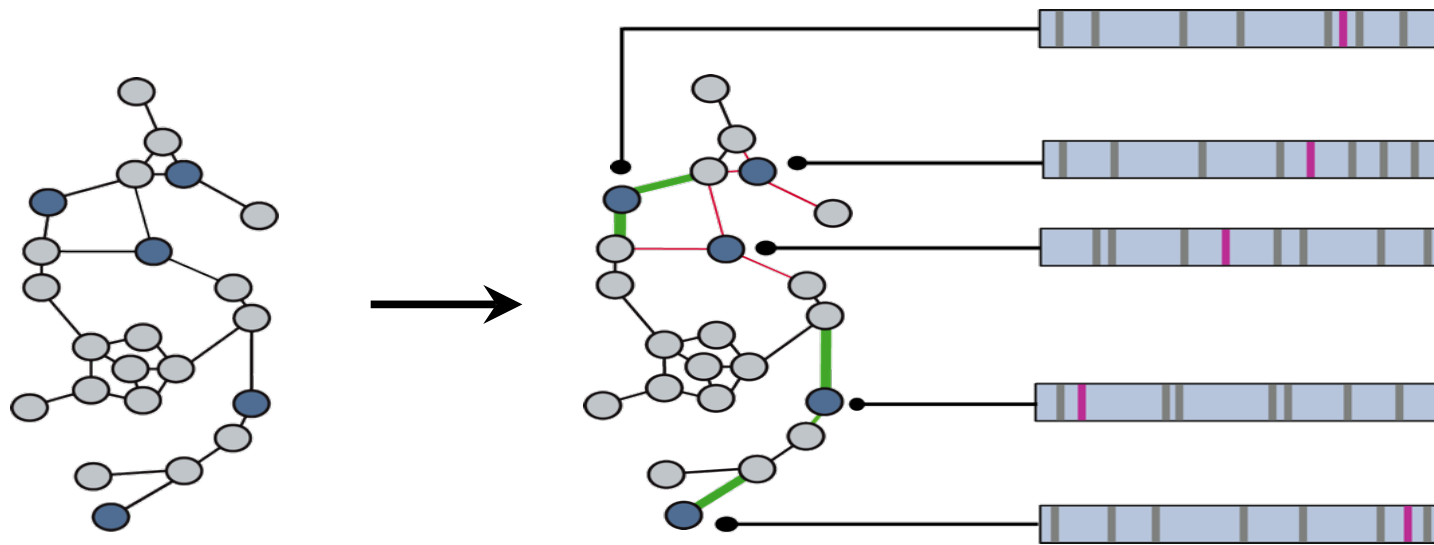


- Variation in protein Interaction networks are a direct and mechanistic projection of environment and genotype onto phenotype
- Genetic variation can result in gene expression variation.
- Correlation of gene expression variation can provide insight into origins of traits, diseases and their variations.

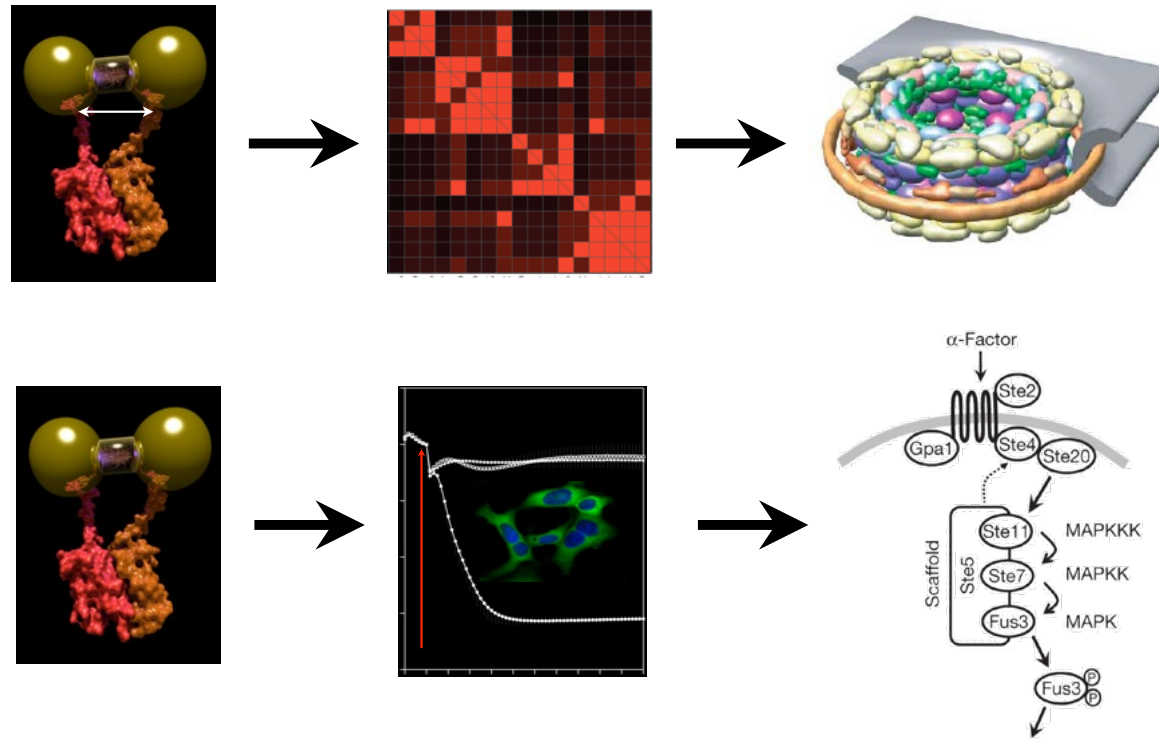
PPIs captures integrated effects of environment on cell state



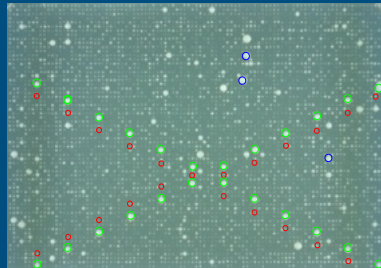
Dynamic PPI Networks identify molecular mechanisms resulting from variation in environment and quantitative trait loci



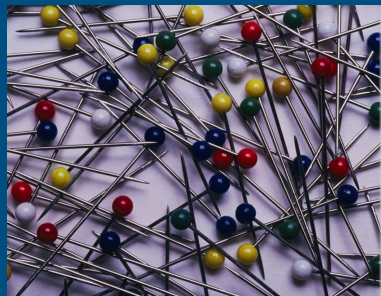
A Molecular Spectrometer of the Cell



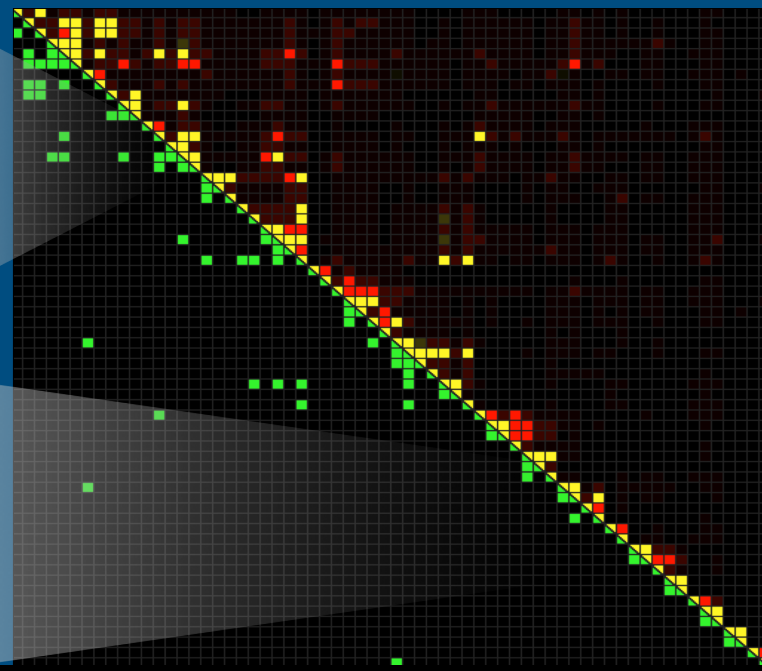
Tarassov *et al. Science*, 2008
 Malleshaiah, *et al. Nature*, 2010



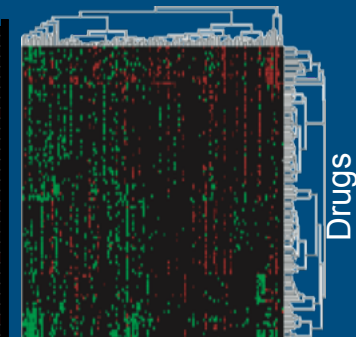
Tarassov et al., *Science*, 2008



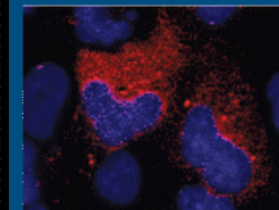
PINs and NeedLs



PCAs



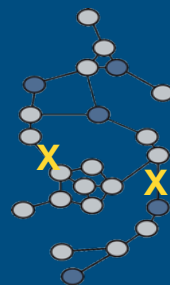
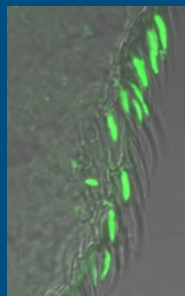
MacDonald et al., *Nat. Chem. Biol.*, 2006
Stynen, et al., *Cell*, 2018



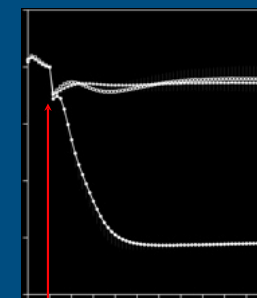
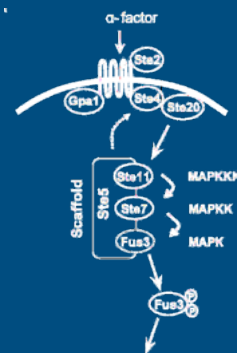
Tchekanda, et al., *Nat. Methods*, 2014



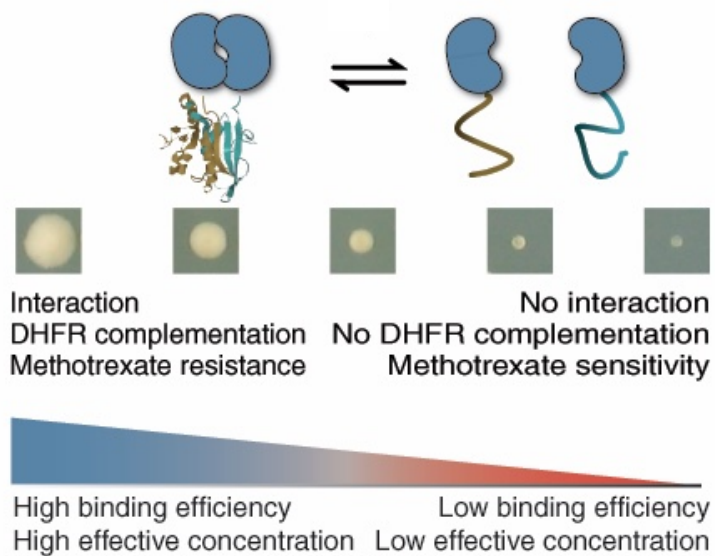
Benton, et al., *PLoS Biol.*, 2006



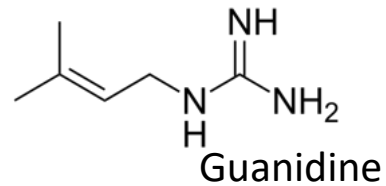
Ear & Michnick,
Nat. Methods, 2009



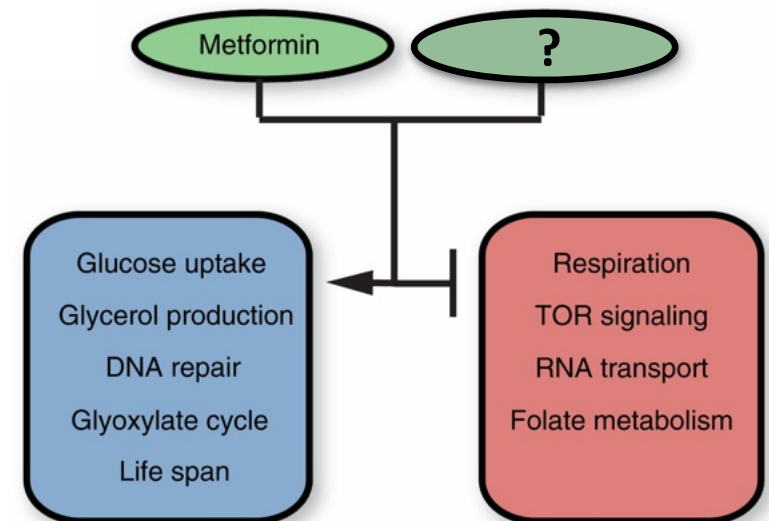
Remy et al., *Nat. Methods*, 2006
Stefan et al., *PNAS*, 2007
Malleshaiah, et al., *Nature*, 2010



Galegin

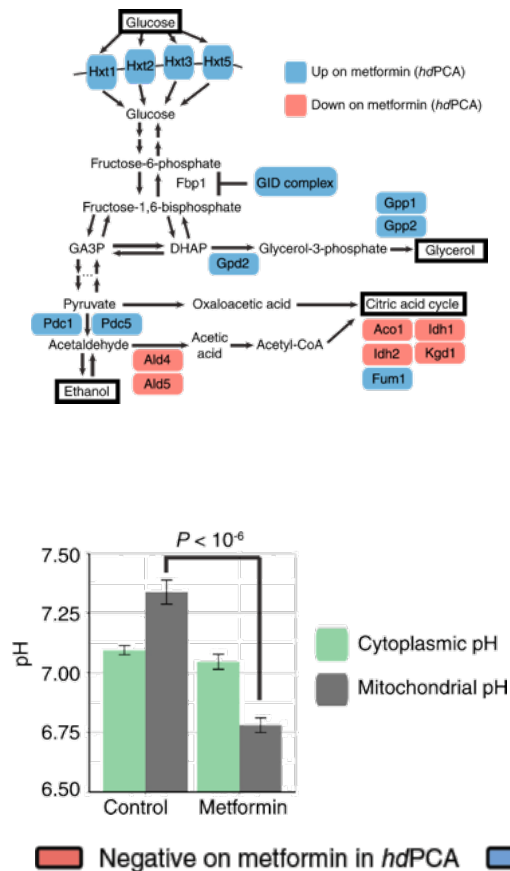


Gerardo Ferbeyre

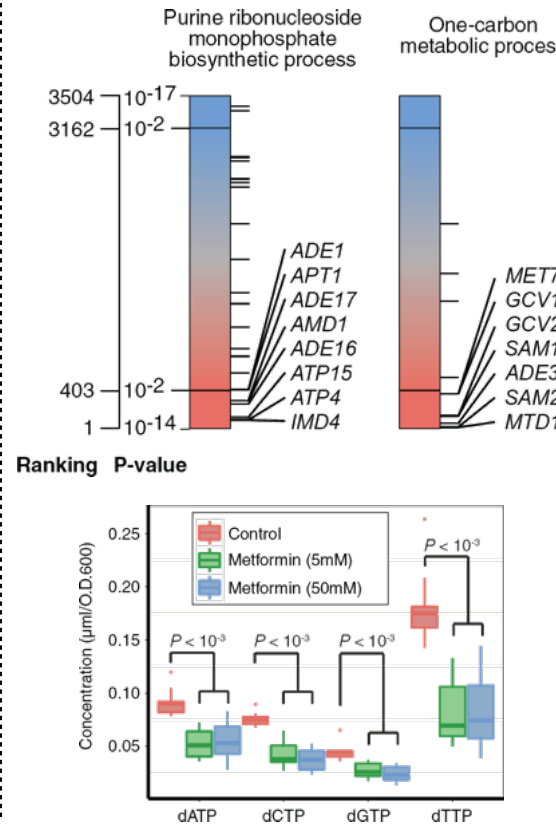


Metformin-targeted processes inferred by *hdPCA*

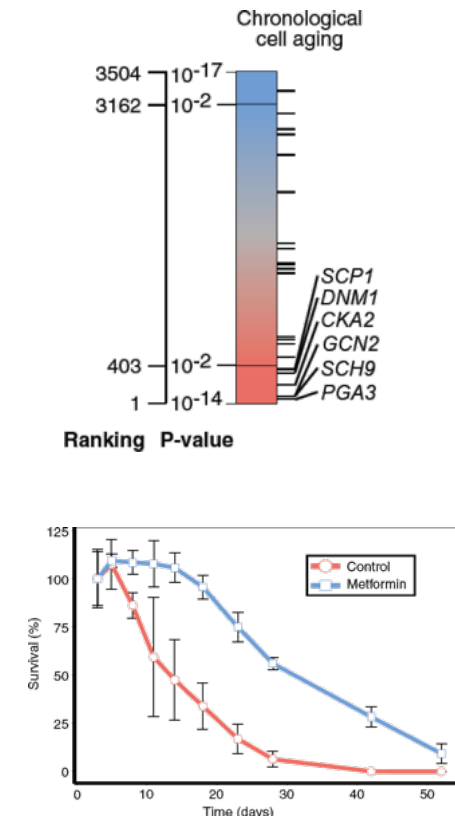
Complex I



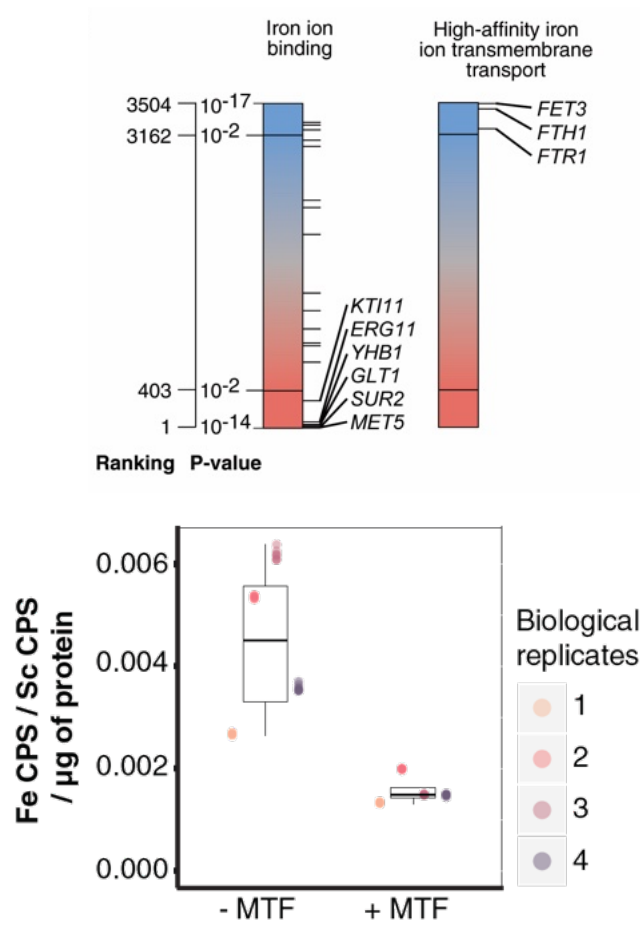
Nucleotide synthesis



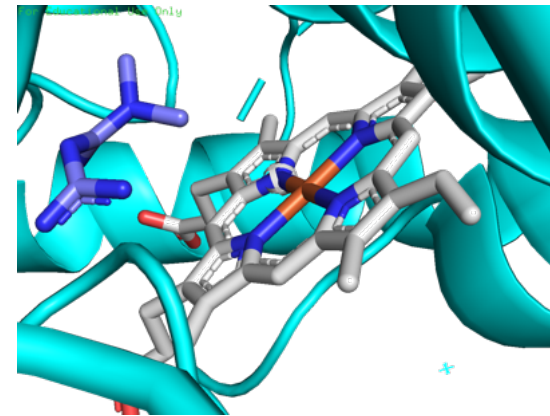
Chronological aging



Metformin induces an iron deficiency-like state



Stylen, et al., *Cell*, 2018



Heme Binding Biguanides Target Cytochrome P450-Dependent Cancer Cell Mitochondria

Guo, Z., et al., *Cell Chem. Biol.*, 2017

Metformin also selectively kills cancer persister/stem cells

Hirsch, et al., *Cancer Res.*, 2009; Hangauer, et al., *Nature.*, 2018

Persister cancer cells, iron addiction and vulnerability to ferroptosis

Rodriguez, et al., *Mol. Cell*, 2022

Metformin Induces ferroptosis in breast cancer

Yang, J., et al., *J. Exptl. & Clin. Canc. Res.*, 2022

The genetic landscape of an in vivo protein interactome in *Saccharomyces cerevisiae*

Besse S, Sakaguchi T, *et al.* *bioRxiv* 2023.12.14.571726

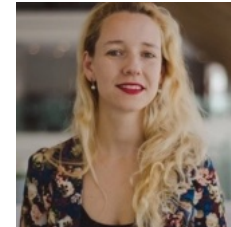
Tatsuya Sakaguchi



Savandara Besse

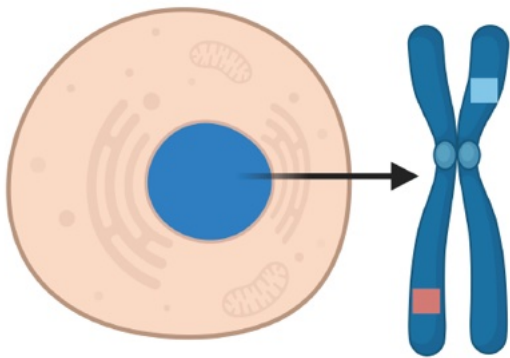


Julie Hussin (Montreal Heart Institute)

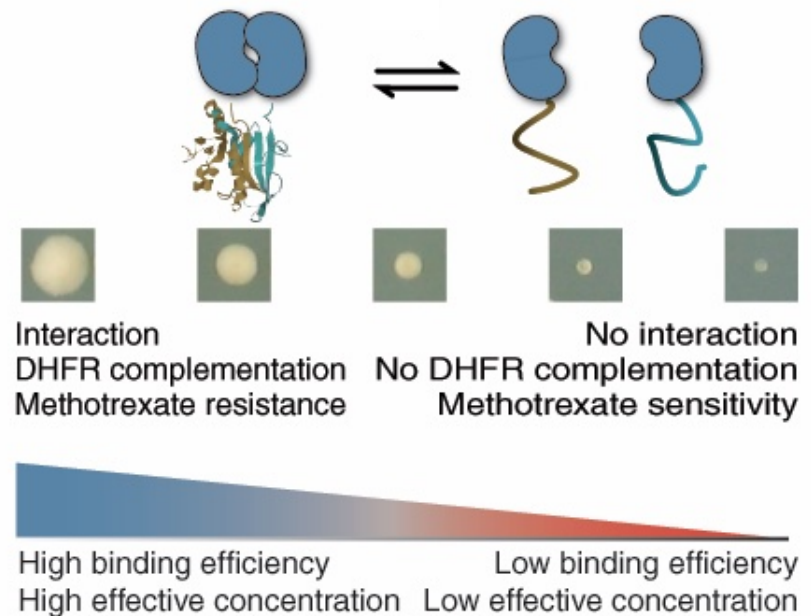
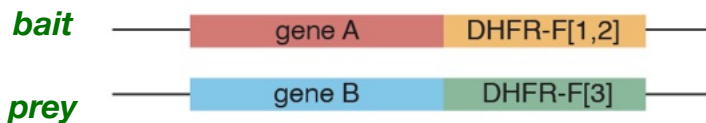


Measuring *in vivo* protein-protein interaction by DHFR PCA

Protein-fragment Complementation Assay (PCA) based on the enzyme **Dihydrofolate Reductase (DHFR)**

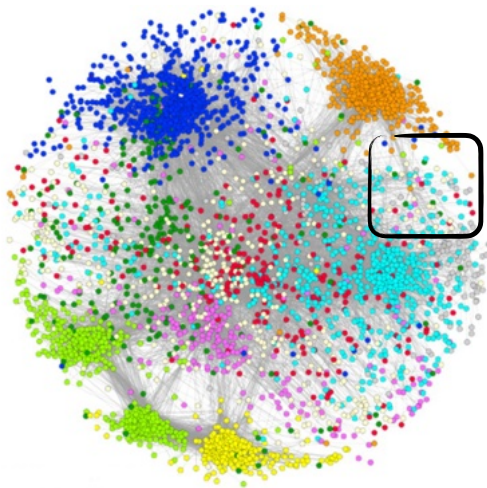


Tag protein pair with mrDHFR fragments

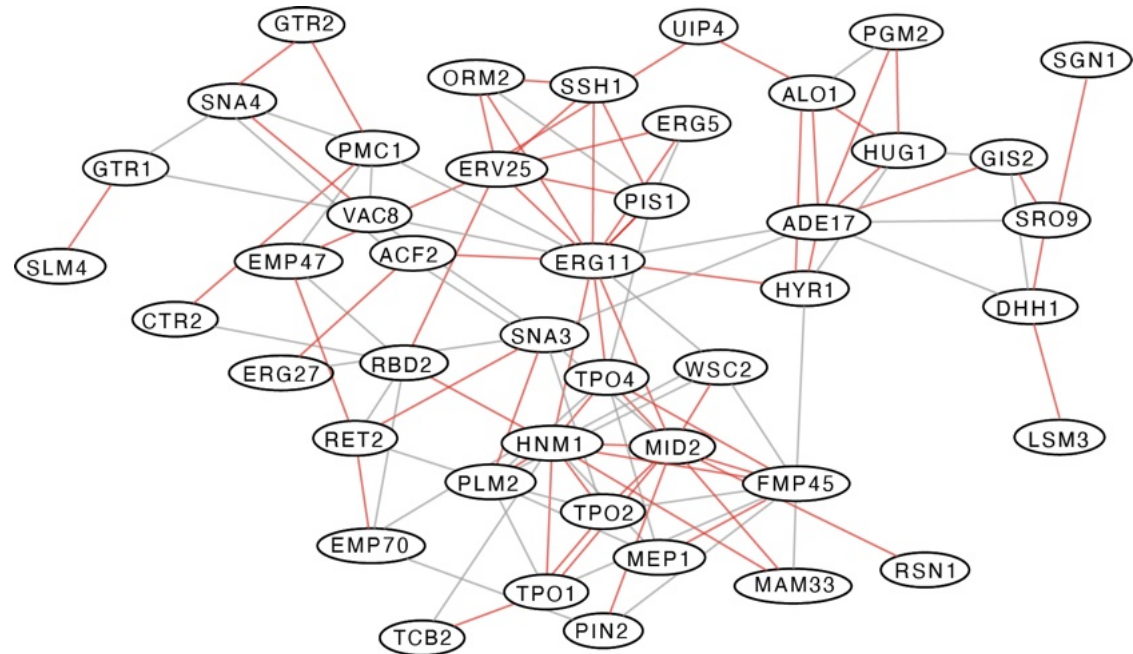


Tarassov, *et al.*, *Science*, 2008
Stynen *et al.*, *Cell* 2018

The “Small World” of PPI Networks Allows Us to Probe it with a PPI Sub-network

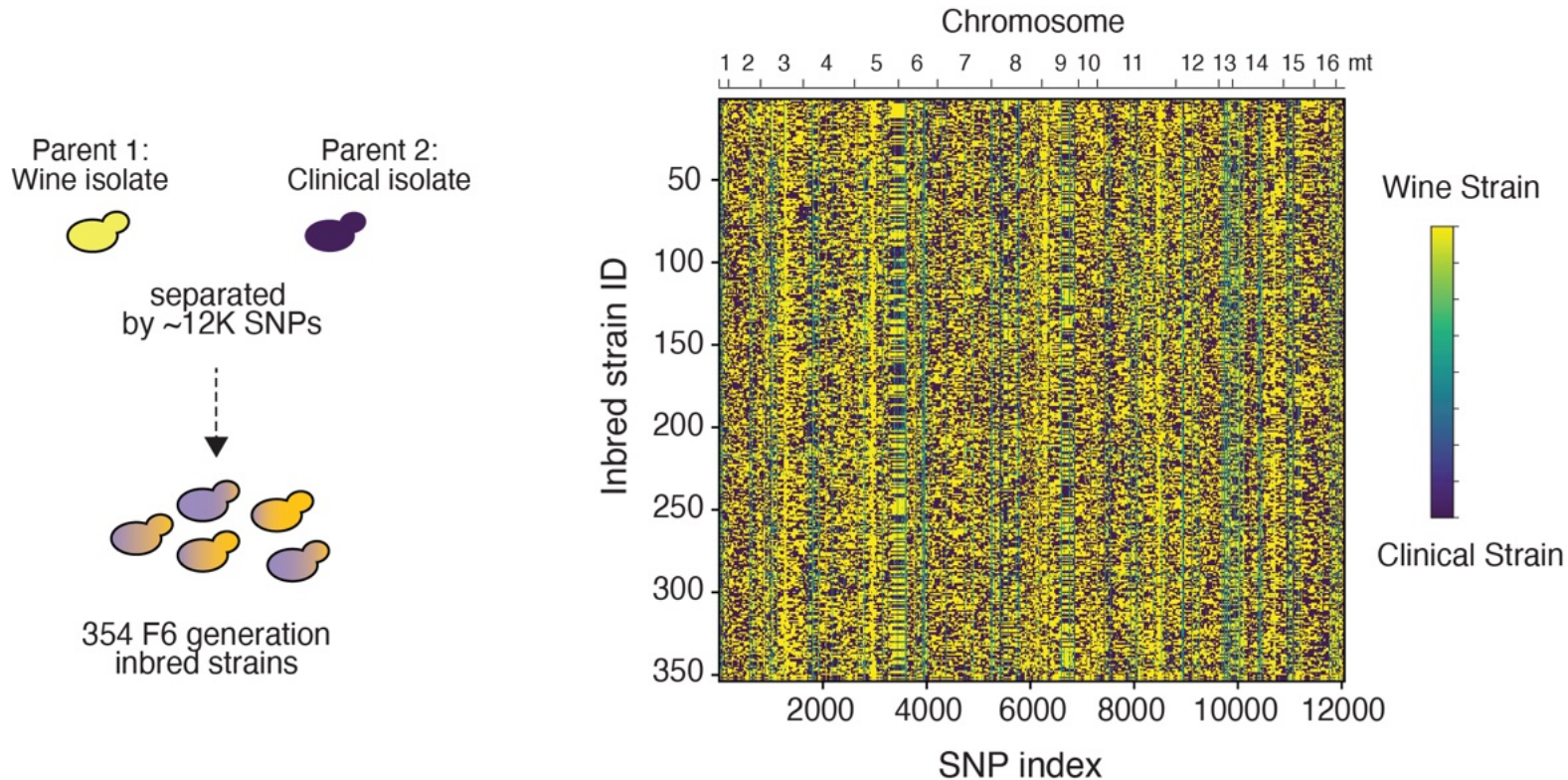


Protein-protein interaction
network (STRING)



61 PPIs for proof-of-concept
development of piQTL

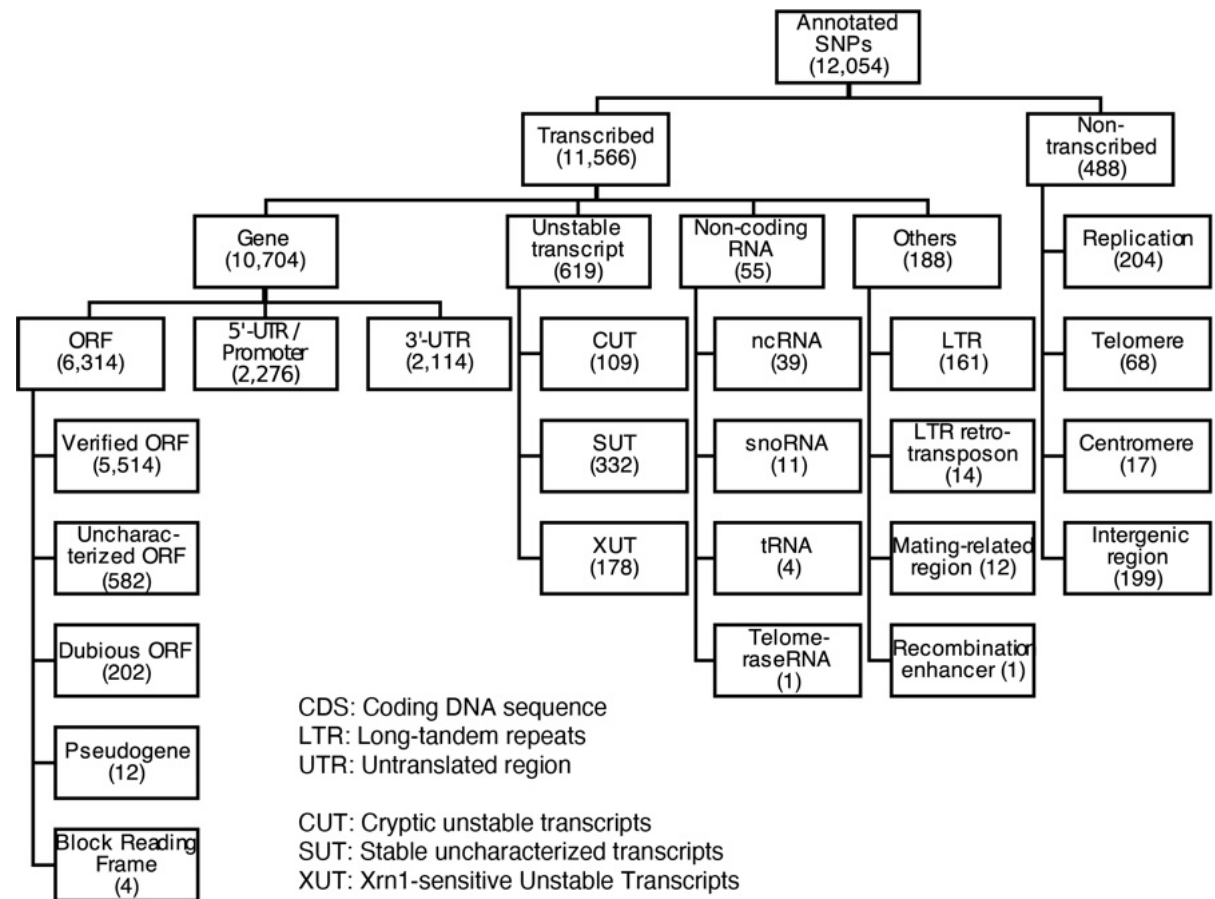
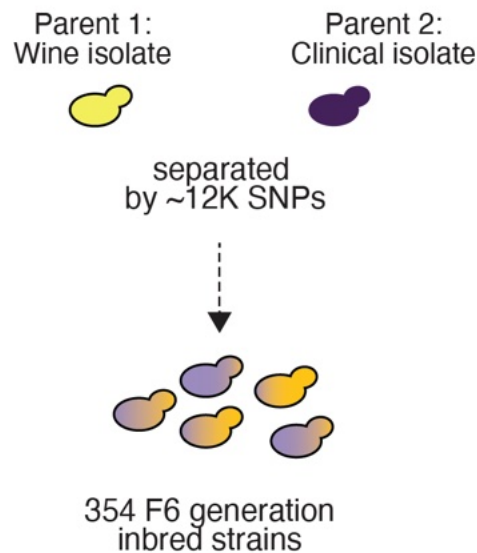
Inbred yeast strains with minimum linkage disequilibrium could provide nucleotide-resolution effect on PPI



~1 SNP per kb

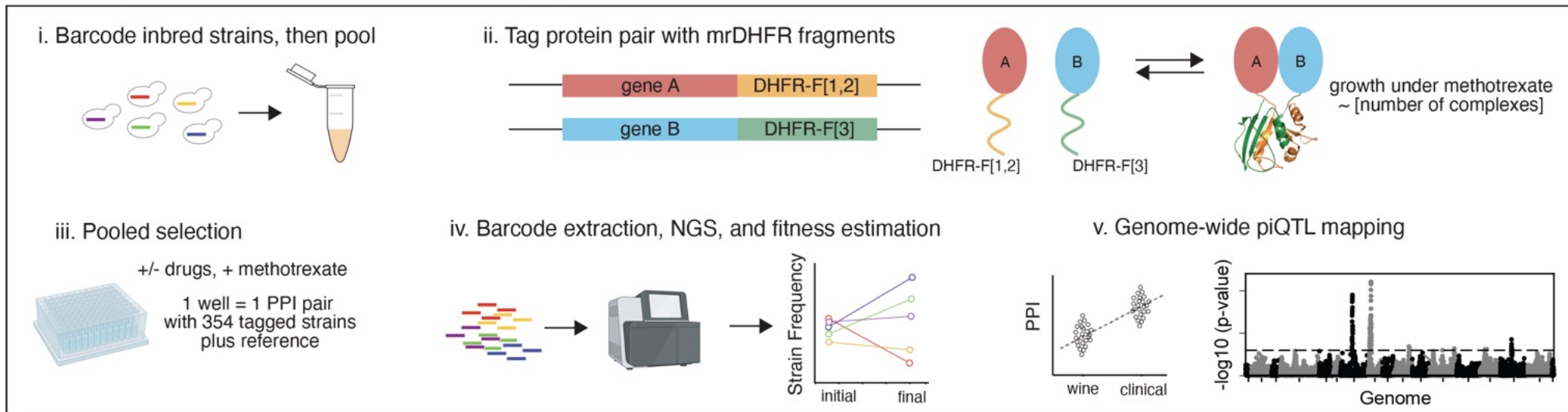
SNPs are uncorrelated (minimal linkage disequilibrium)

SNPs cover diverse genomic elements, including those with unknown function



Multiplexed DHFR PCA for piQTL mapping

In vivo protein interaction QTL (piQTL) mapping



Genotype X Environment = Molecular Phenotype

**353 yeast
strains**

5 drugs

**61 protein-protein
interactions**

Selection for PPI strength

- Methotrexate

Drugs relevant to yeast

- Fluconazole (antifungal)
- 5-fluorocytosine (5-FC) (antifungal)

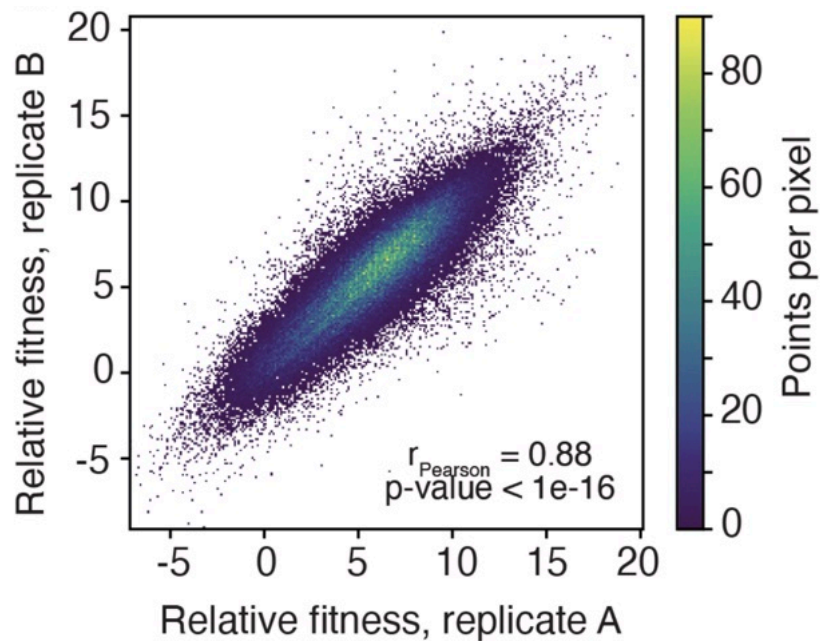
Drugs relevant to humans

- Metformin (diabetes)
- Trifluoperazine (schizophrenia)

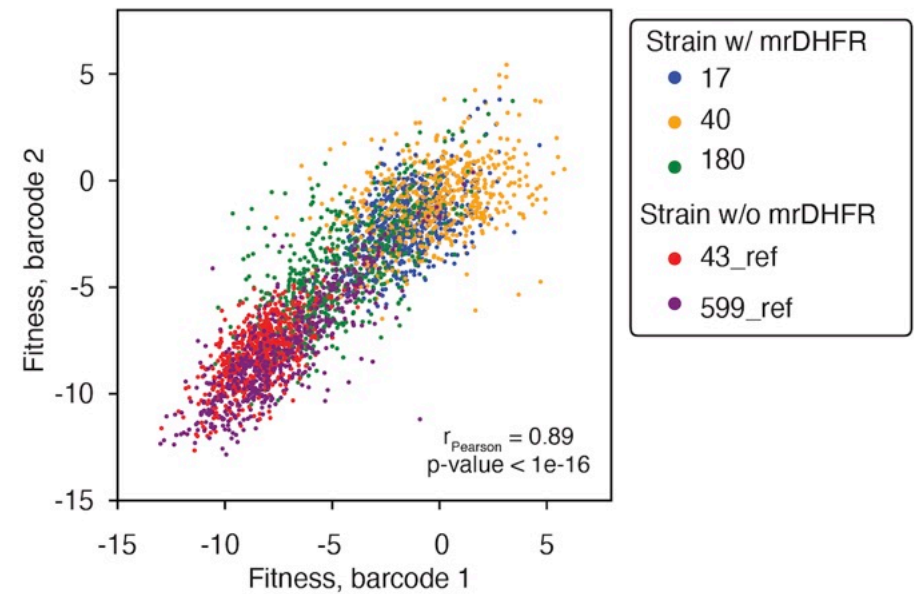
**~1,000,000 growth
measurements**

Fitness estimates are reproducible in the screen

Biological replicates

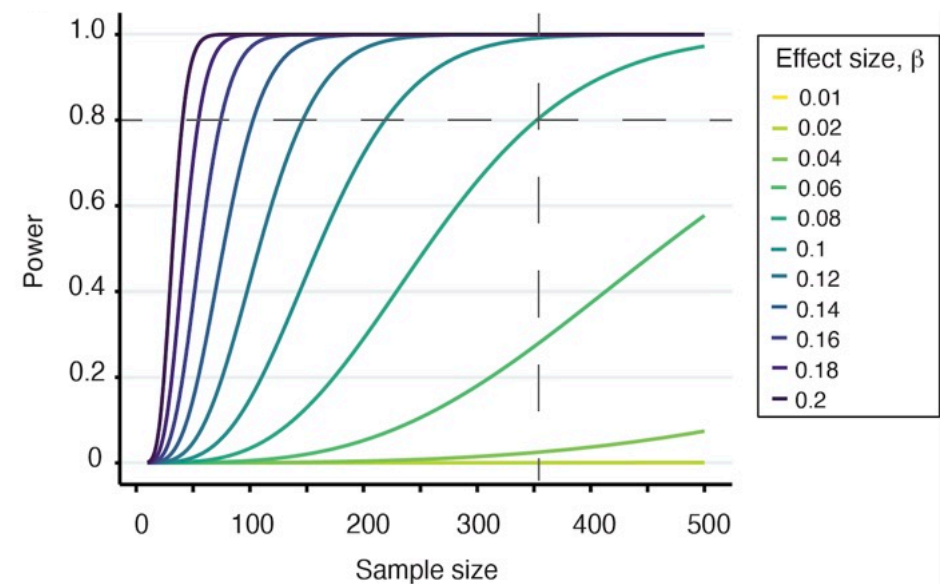
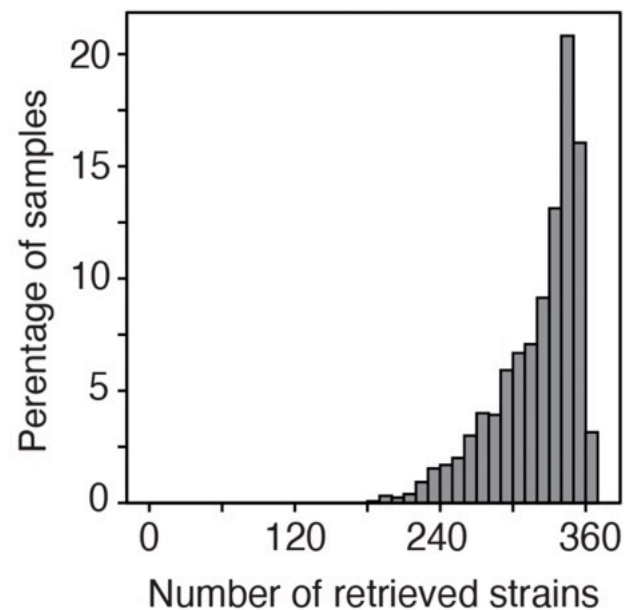


Within-well (“internal”) controls

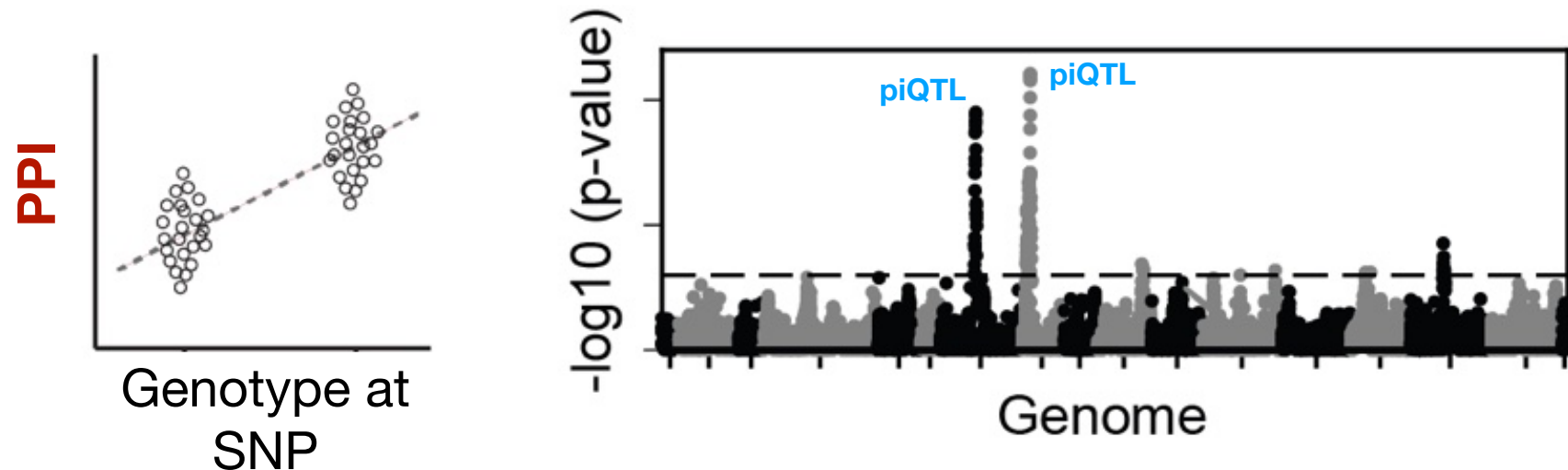


Multiplex DHFR PCA screen recovers >95% of the 353 genomes

Power calculation



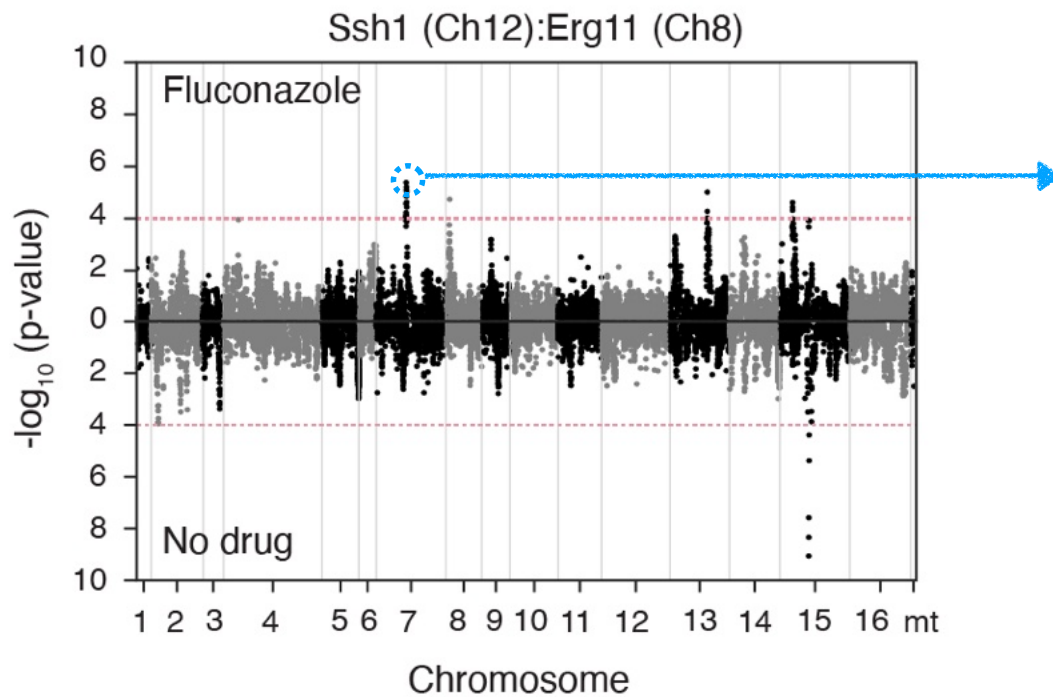
piQTL mapping follows the same logic of other QTL analyses



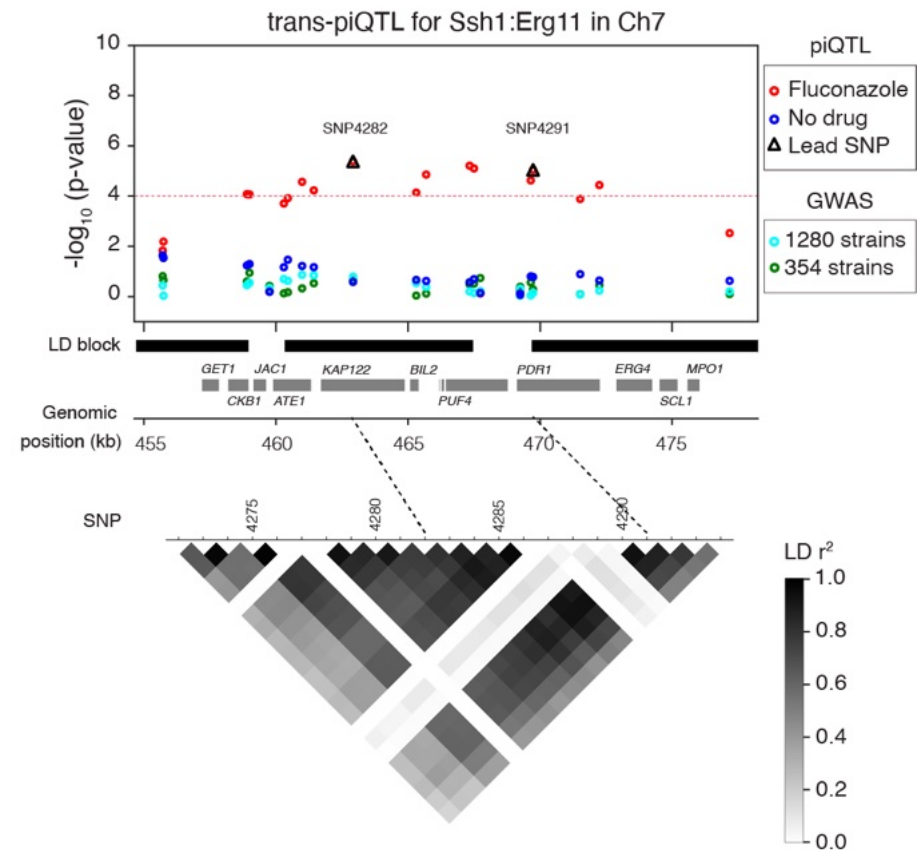
$$PPI = \alpha + \beta * Genotype_{SNP} + \epsilon$$

β = “effect size” quantifies the sensitivity of the PPI to the SNP

pi-QTLs capture known drug resistance genes for antifungals



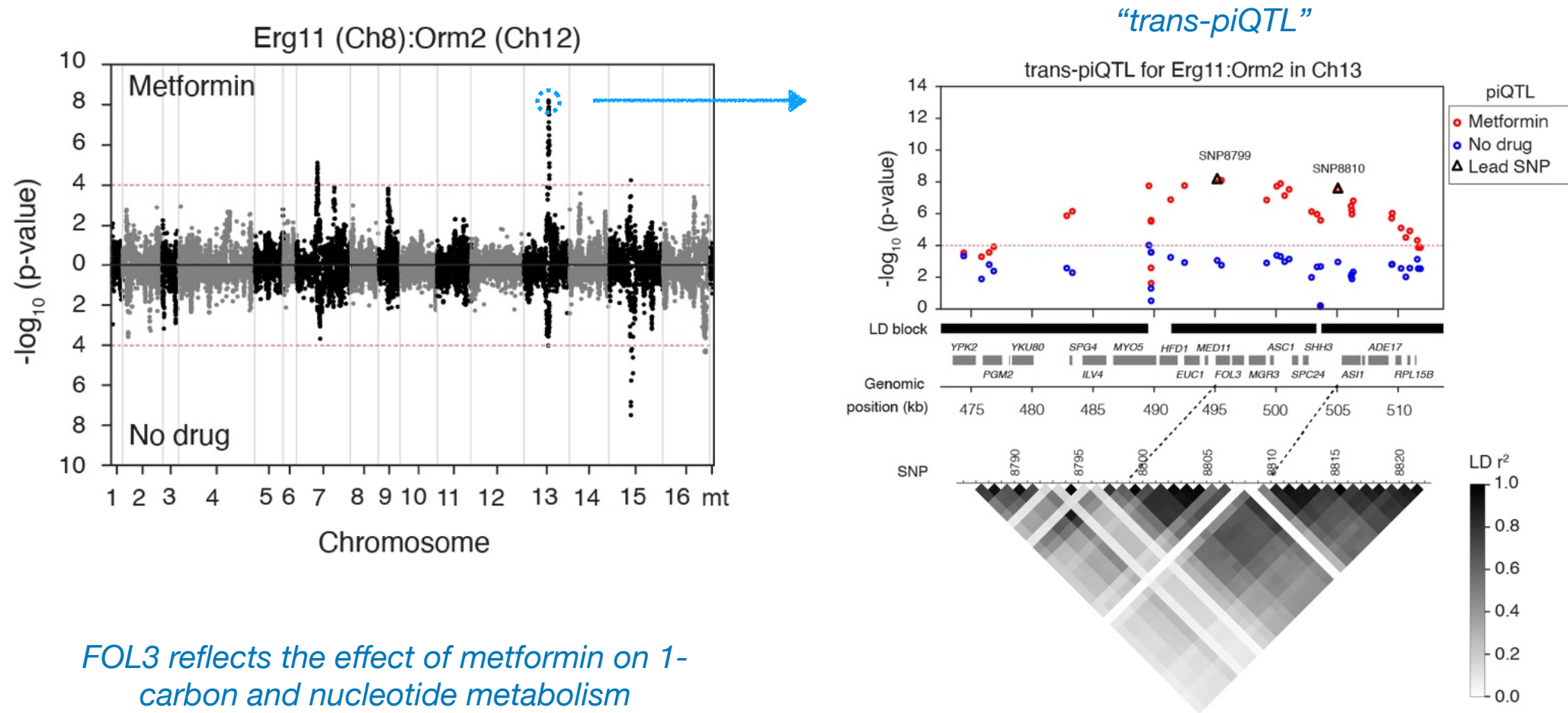
“trans-piQTL”



PDR1 are transcription factors for efflux pumps and transporters

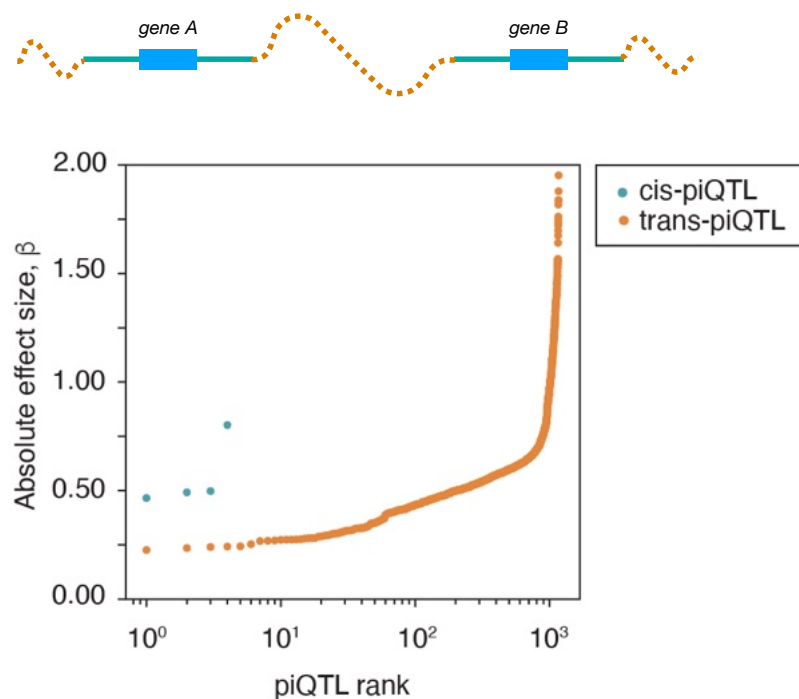
KAP122 (“PDR6”) is nuclear transport protein

pi-QTLs capture known mechanisms of metformin, a type 2 diabetes drug

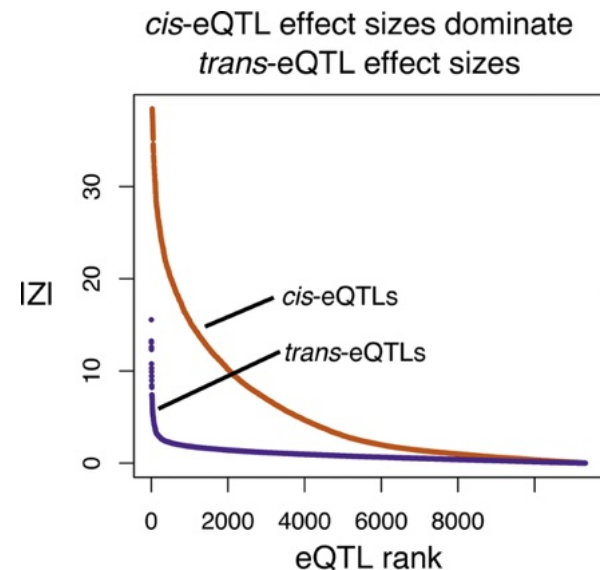


FOL3 reflects the effect of metformin on 1-carbon and nucleotide metabolism

Trans-acting piQTLs are more prevalent and stronger than cis-acting piQTLs, a distinct feature of PPI network



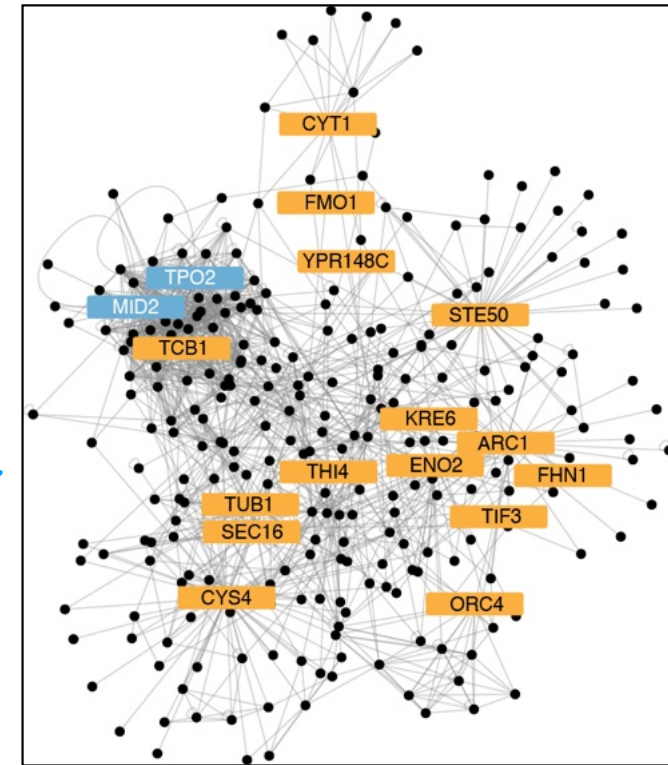
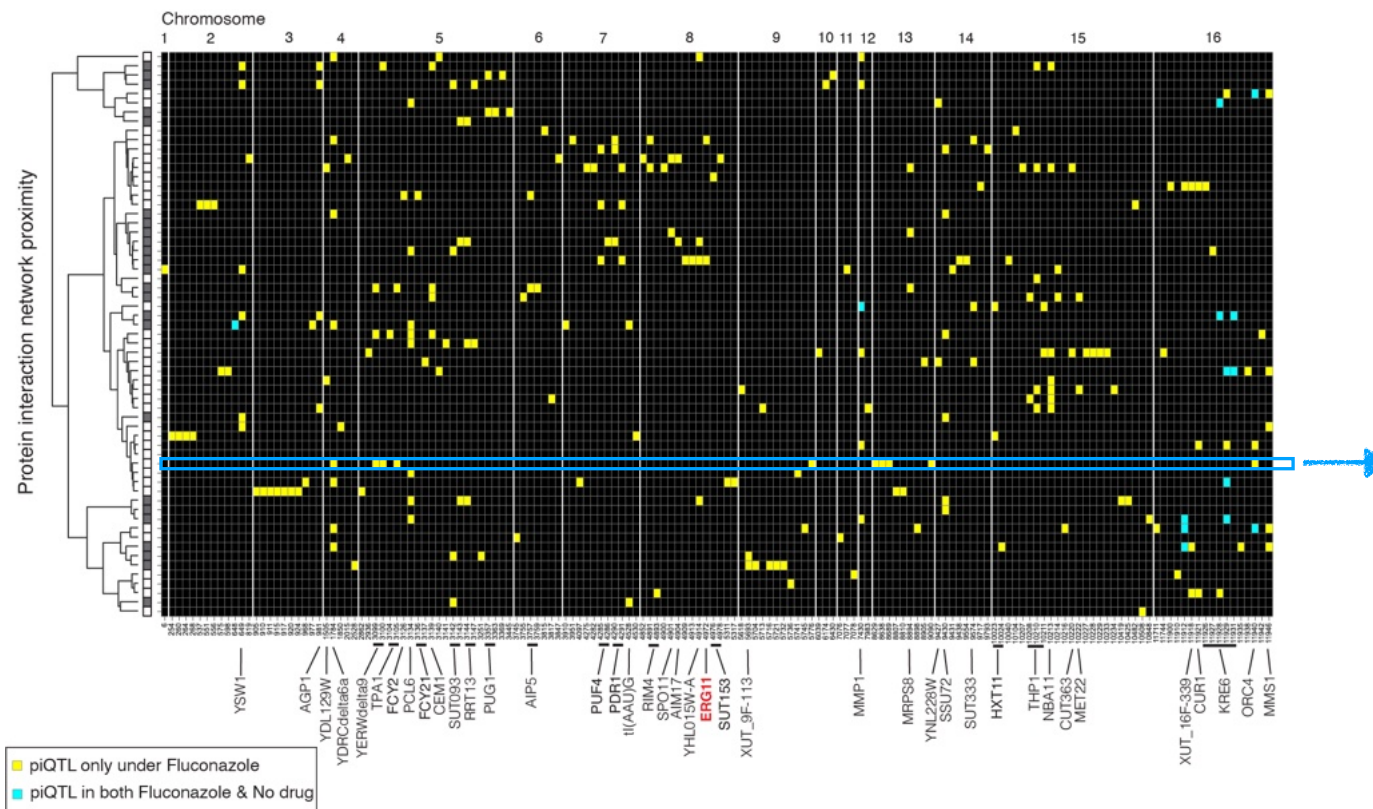
PPI is sensitive to genomic regions far from the gene loci of interacting protein pair.



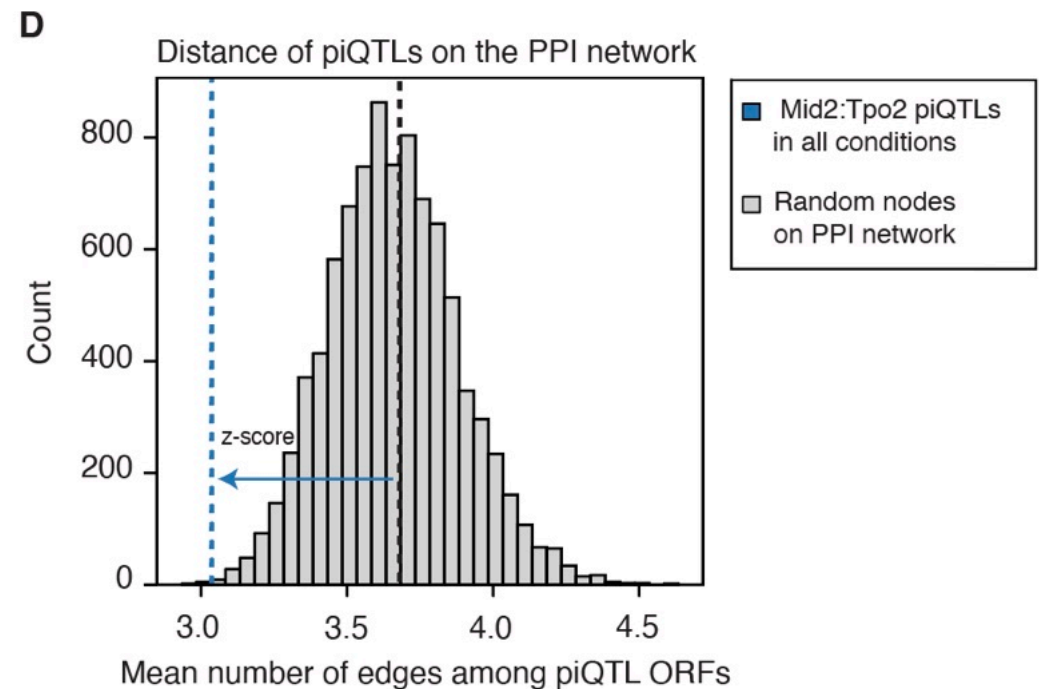
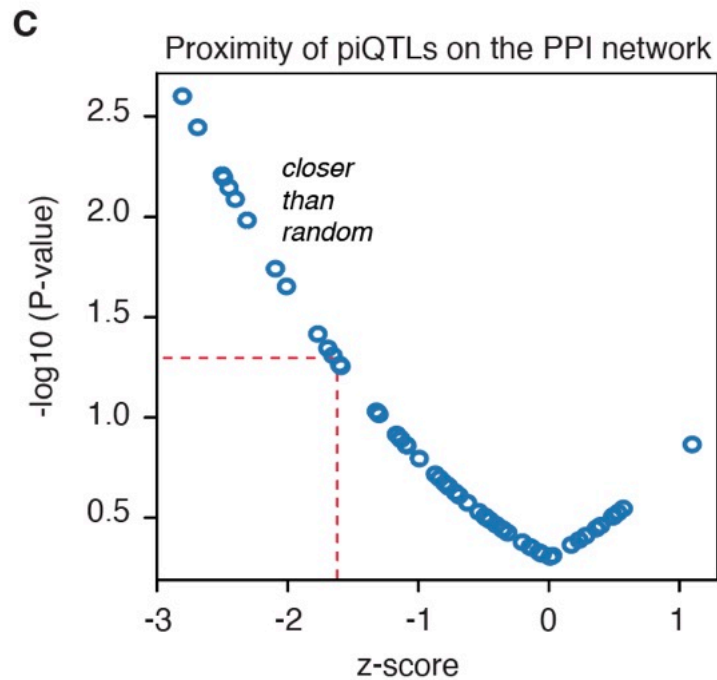
Liu & Pritchard, Cell 2019

In contrast to expression level that is sensitive to SNP in “cis” to the gene.

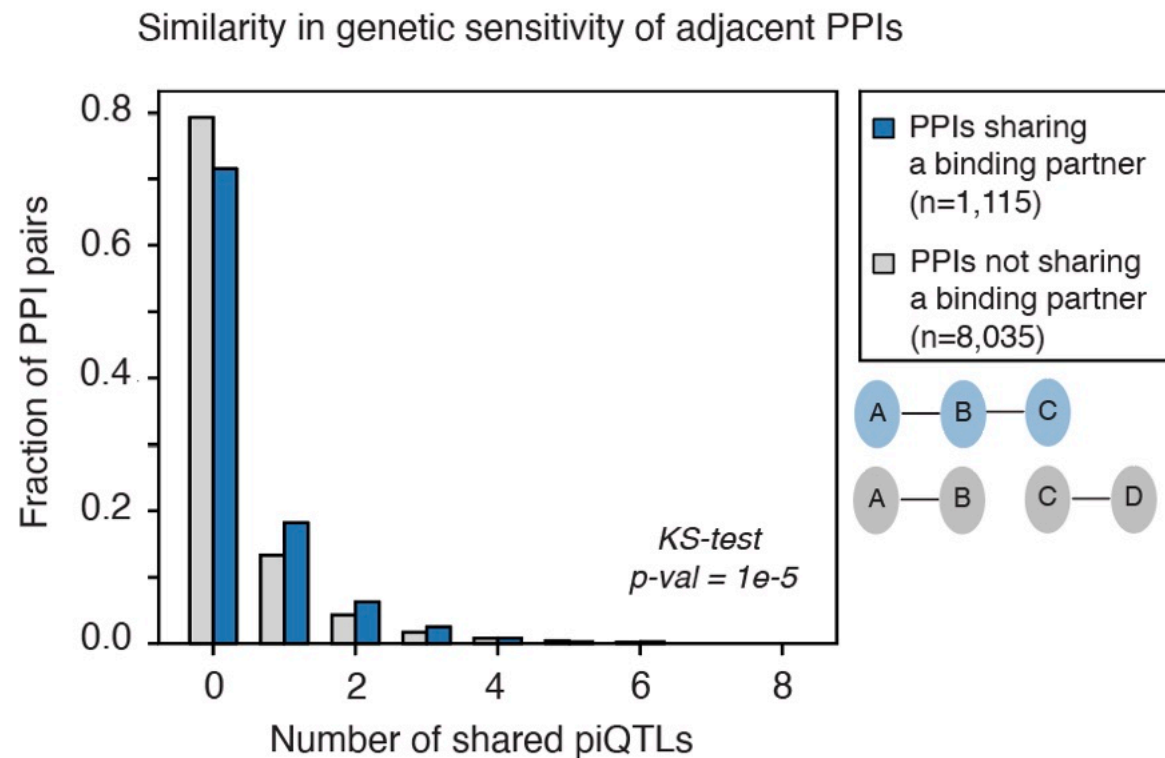
piQTLs are proximal to DHFR PCA reporters demonstrating the small-world nature of the PPI network



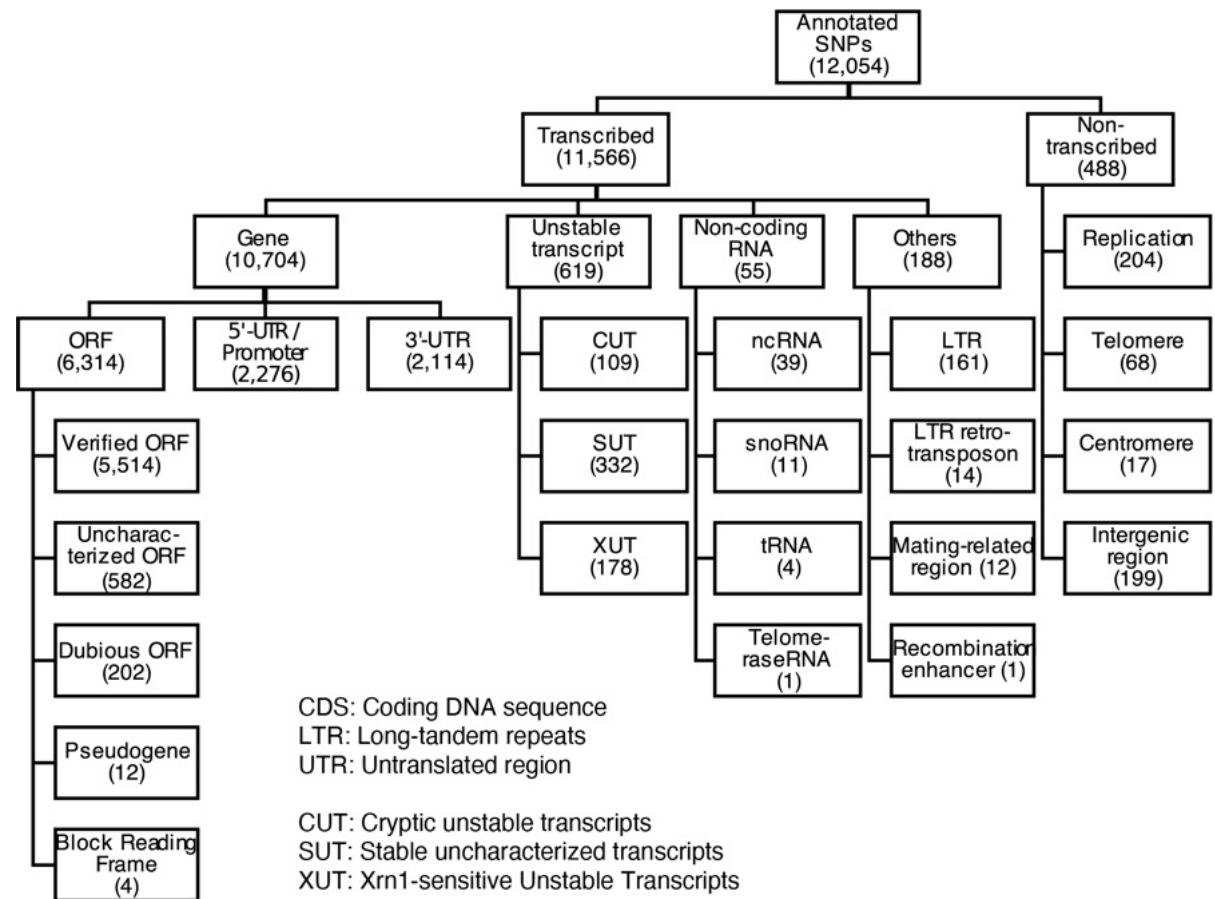
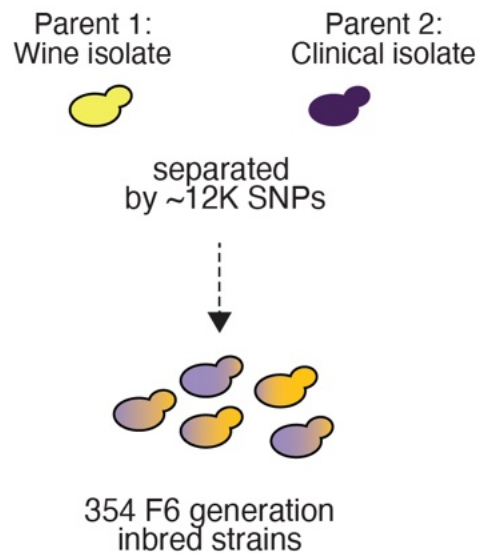
piQTLs are proximal to DHFR PCA reporters demonstrating the small-world nature of the PPI network



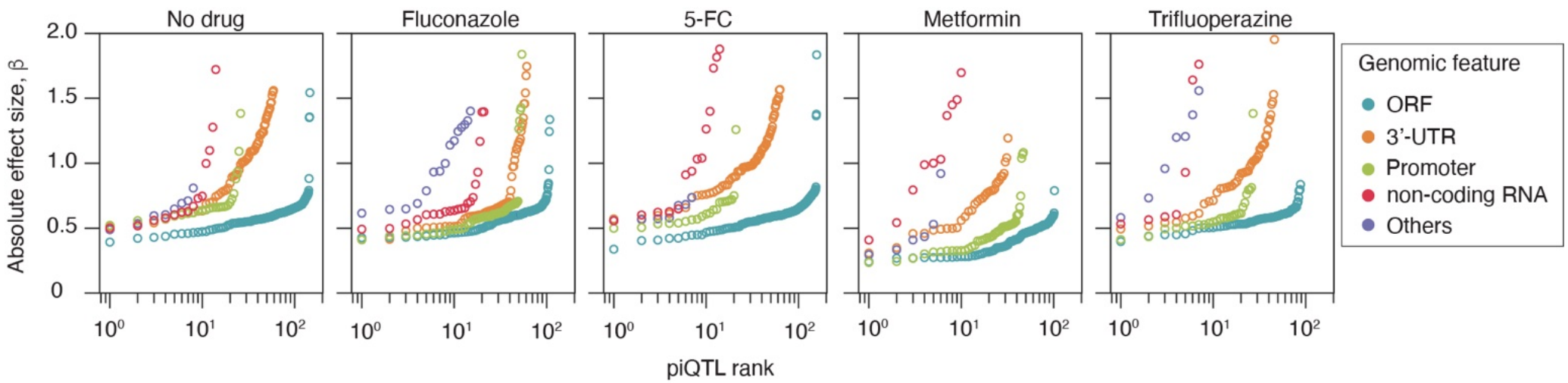
PPIs sharing the same nodes exhibit similar genetic sensitivity



SNPs cover diverse genomic elements, including those with unknown function

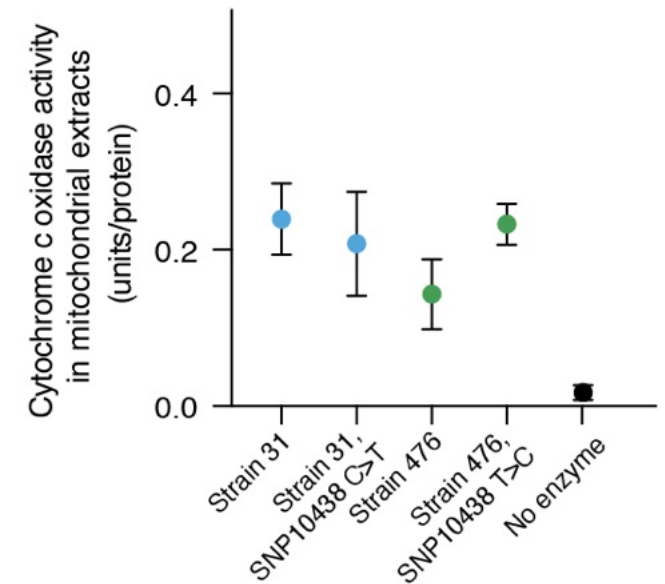
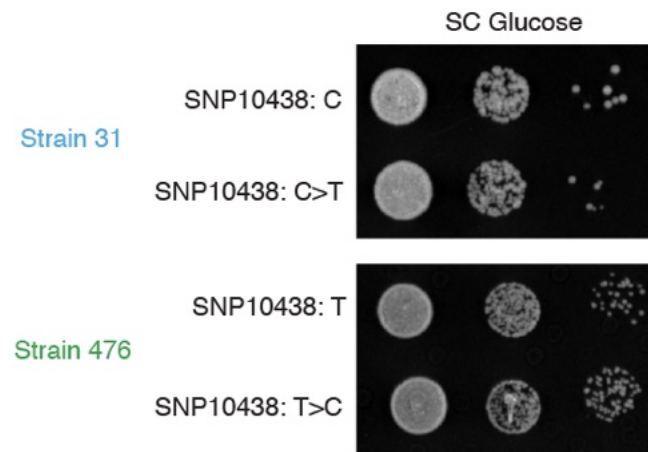
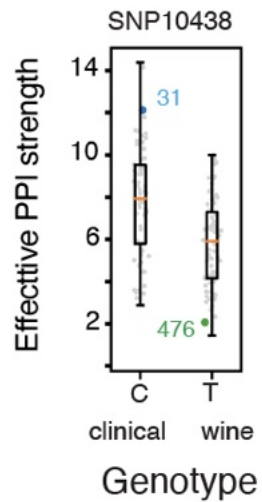
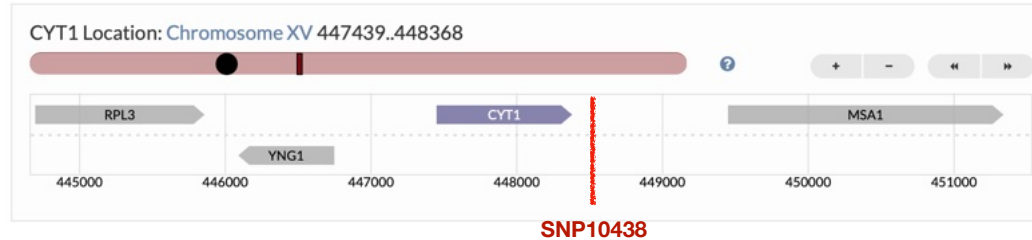


piQTLs are enriched in post-transcriptional regulation and non-coding RNAs

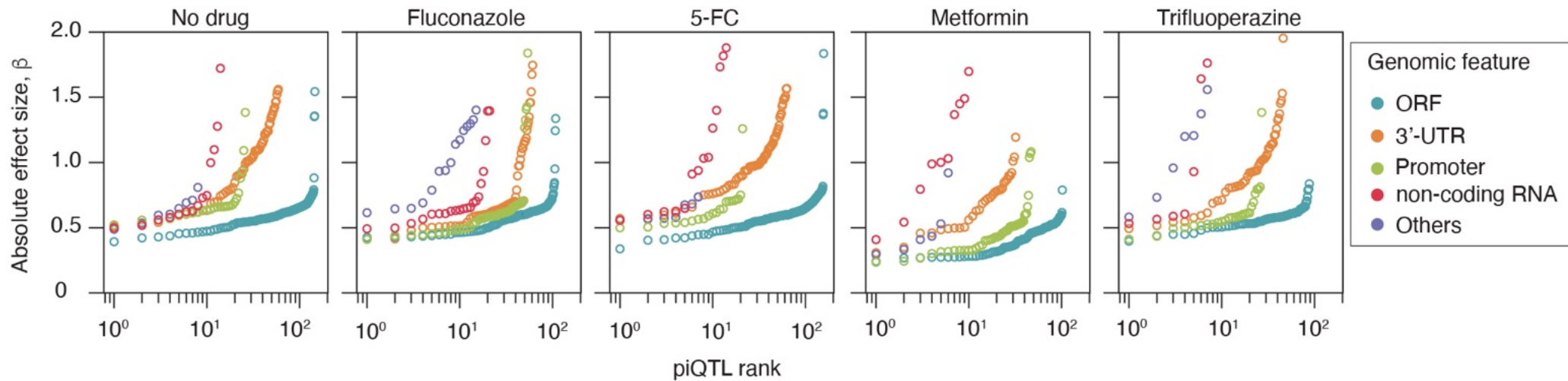


PPIs are strongly sensitive to SNPs in 3'UTR (mRNA localization)

Example: piQTL in the 3'UTR of *Cyt1* (cytochrome c1)

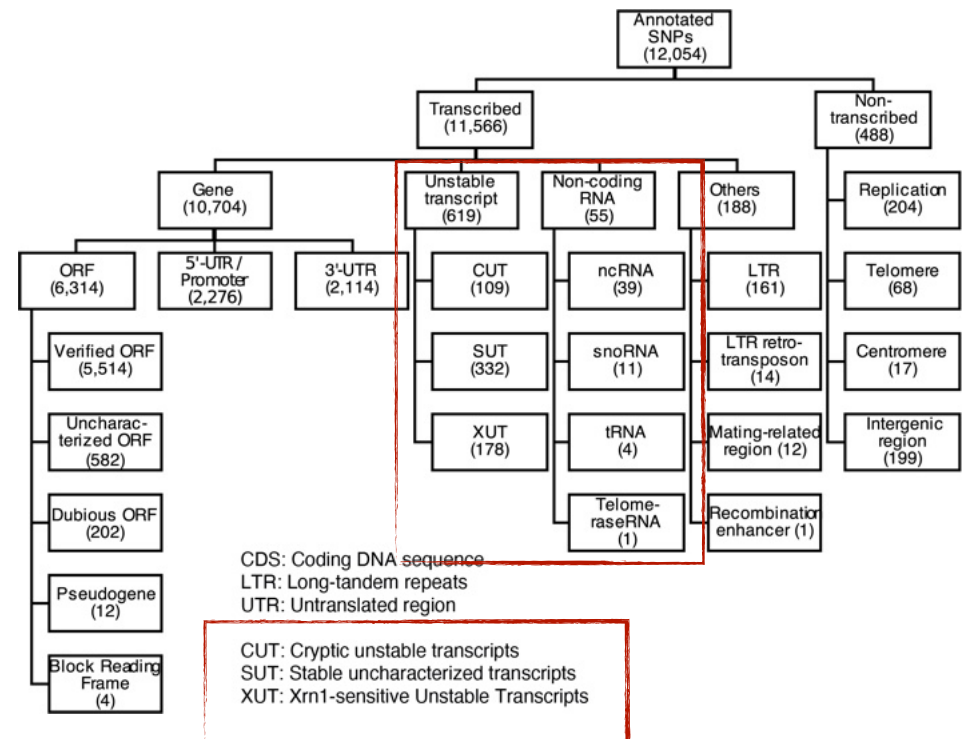
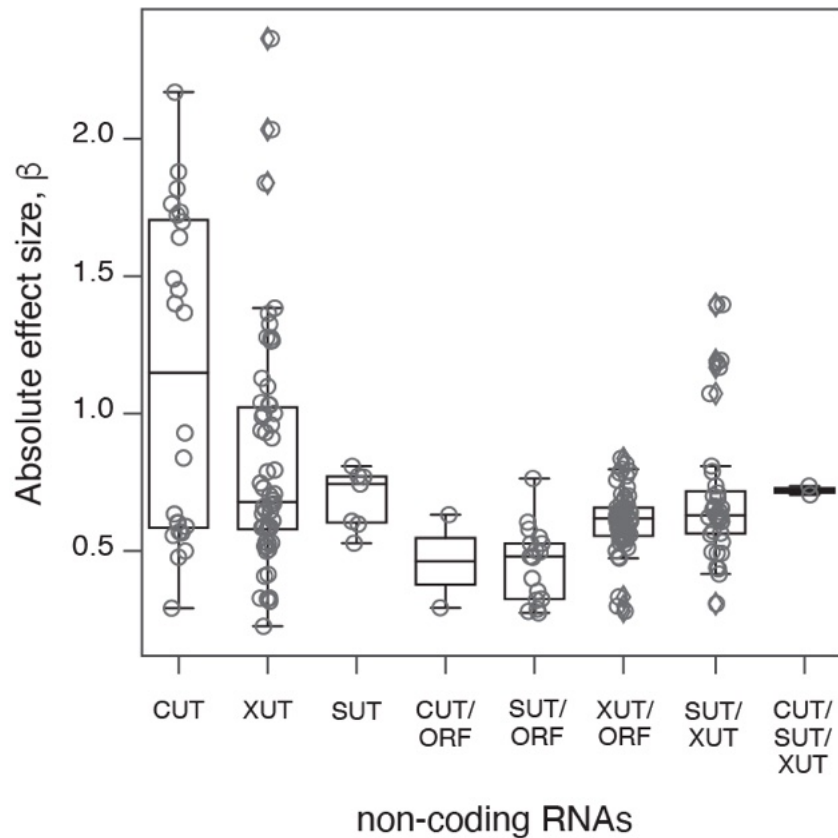


piQTLs are enriched in post-transcriptional regulation and non-coding RNAs



PPI network can be used to probe genome function including functional relevance of ncRNAs.

piQTLs are enriched in non-coding RNAs



CDS: Coding DNA sequence

LTR: Long-tandem repeats

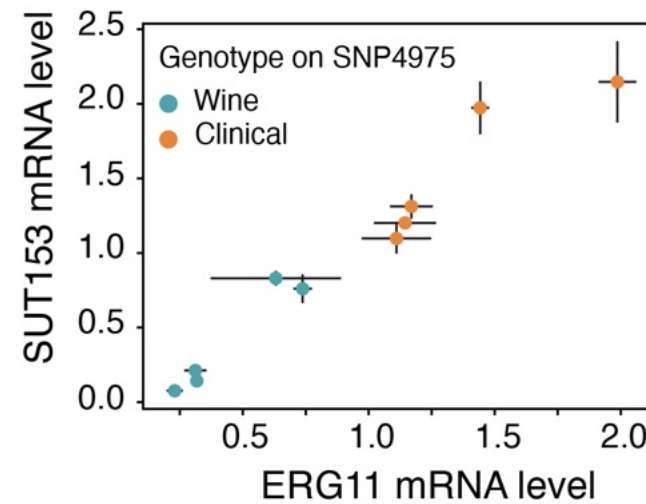
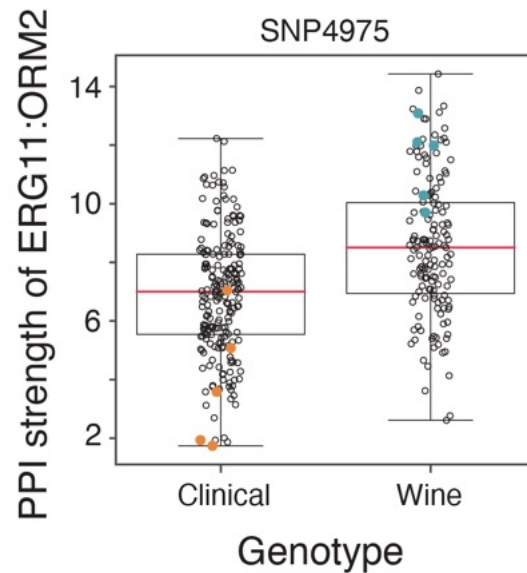
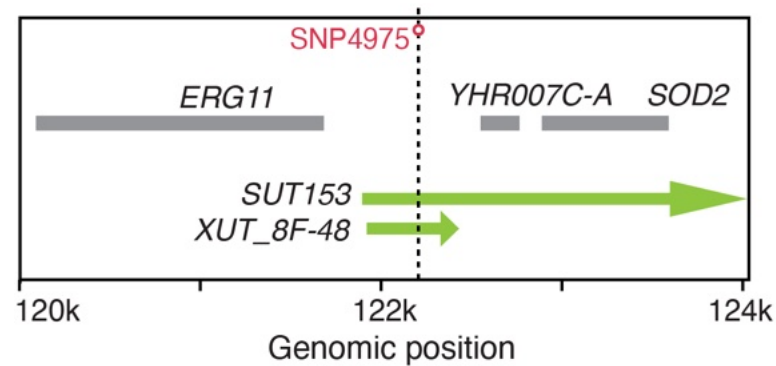
UTR: Untranslated region

CUT: Cryptic unstable transcripts

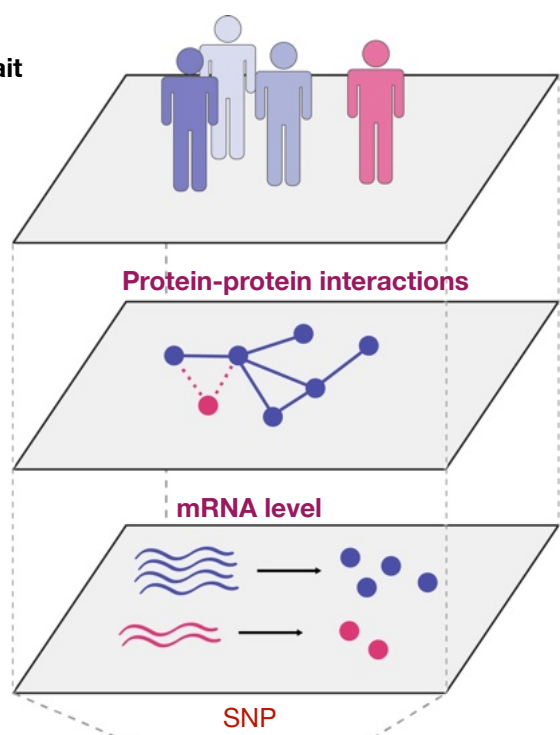
SUT: Stable uncharacterized transcripts

XUT: Xrn1-sensitive Unstable Transcripts

Example piQTLs in non-coding RNAs



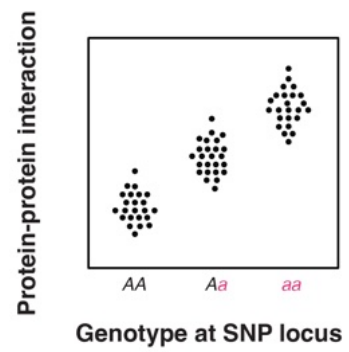
Fitness
Phenotype
Quantitative trait



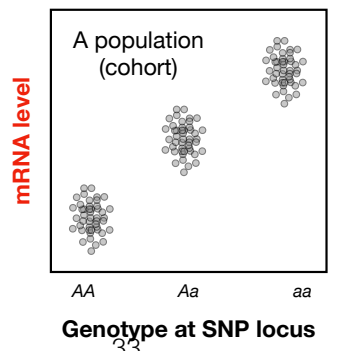
Genomic DNA



piQTL mapping

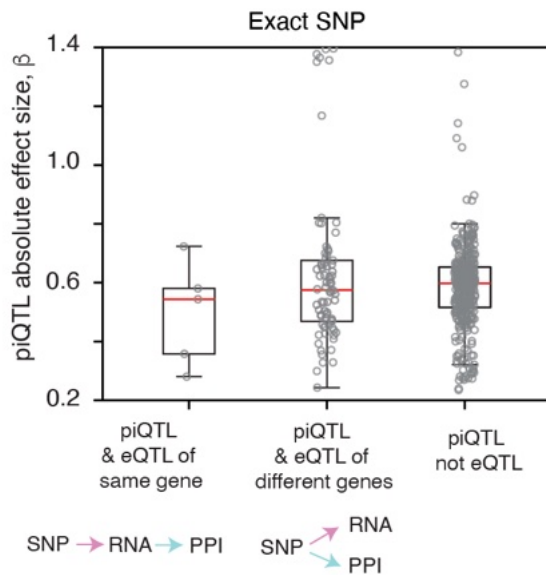
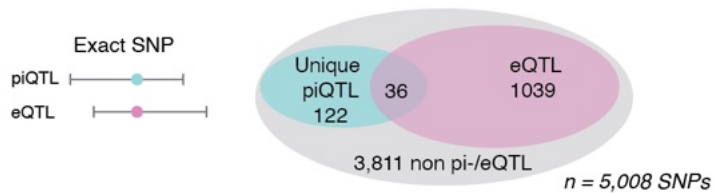


eQTL mapping

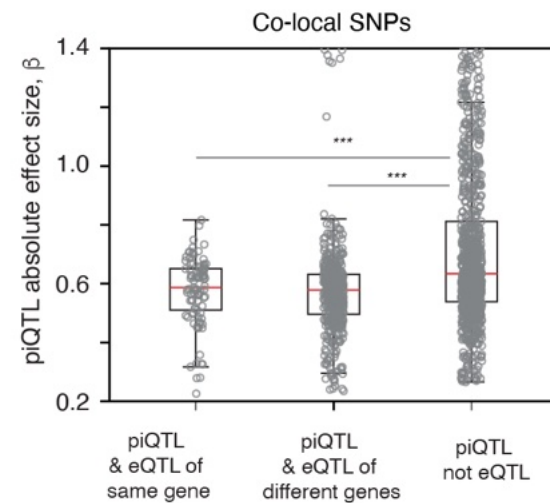
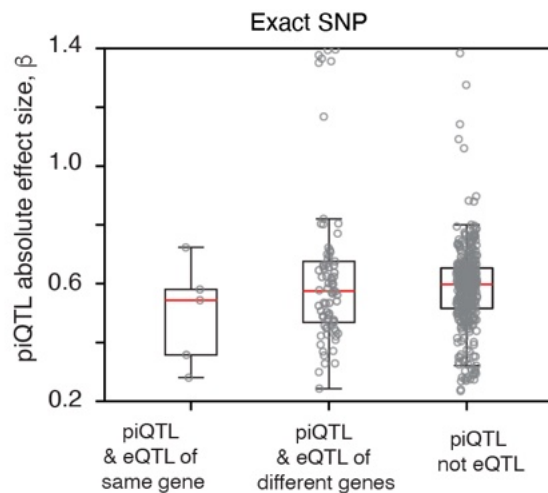
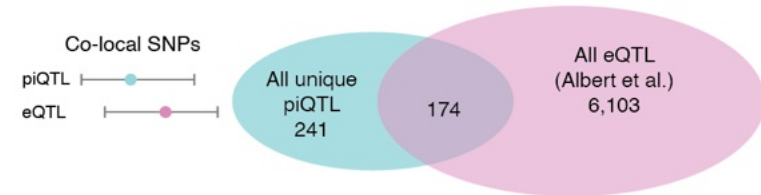
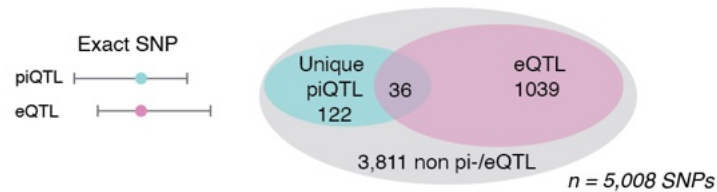


What is the
correspondence between
eQTL and piQTL?

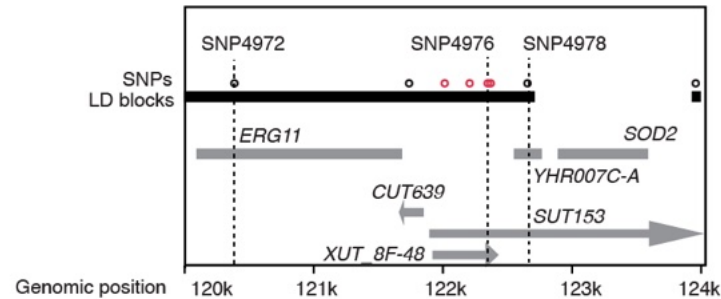
Comparing the effects of genetic variation to transcription and to protein-protein interaction



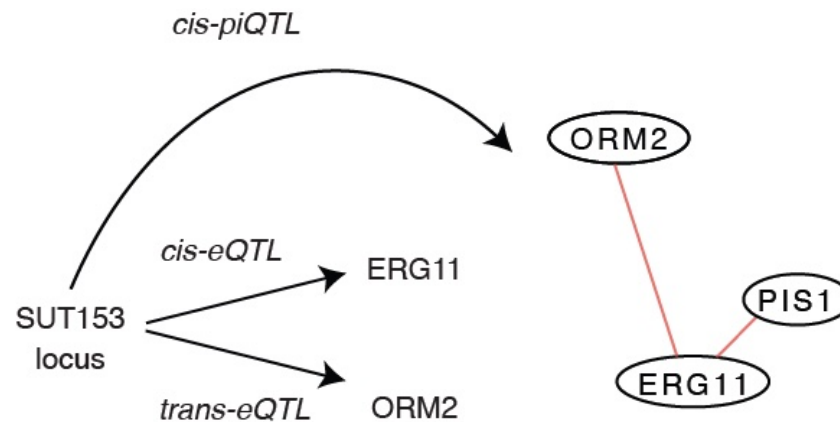
Although there is an overlap of eQTL and piQTL, piQTLs of large effects are not eQTLs



Comparing the effects of genetic variation to transcription and to protein-protein interaction

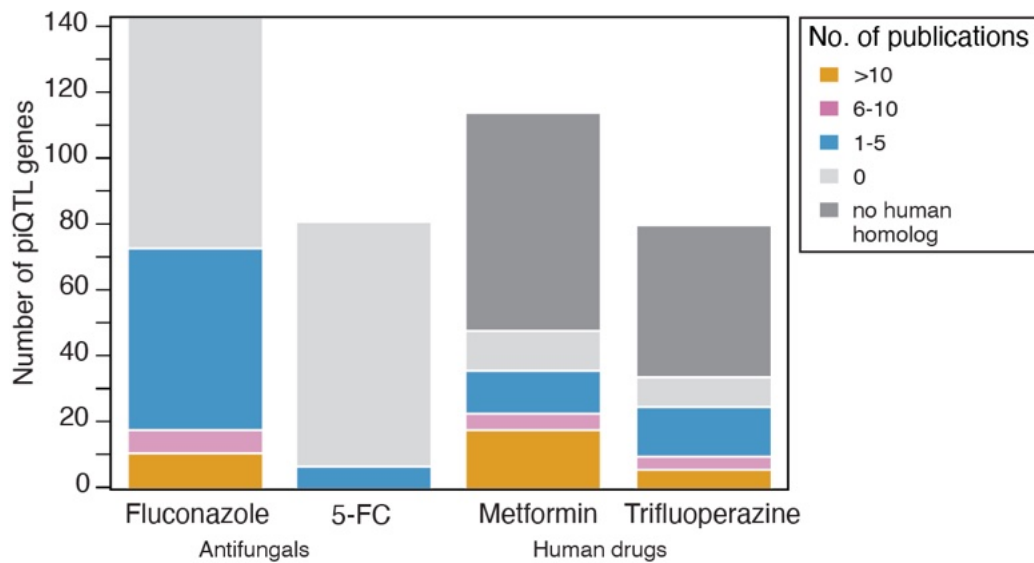


Specific example of a locus that is both an eQTL and piQTL

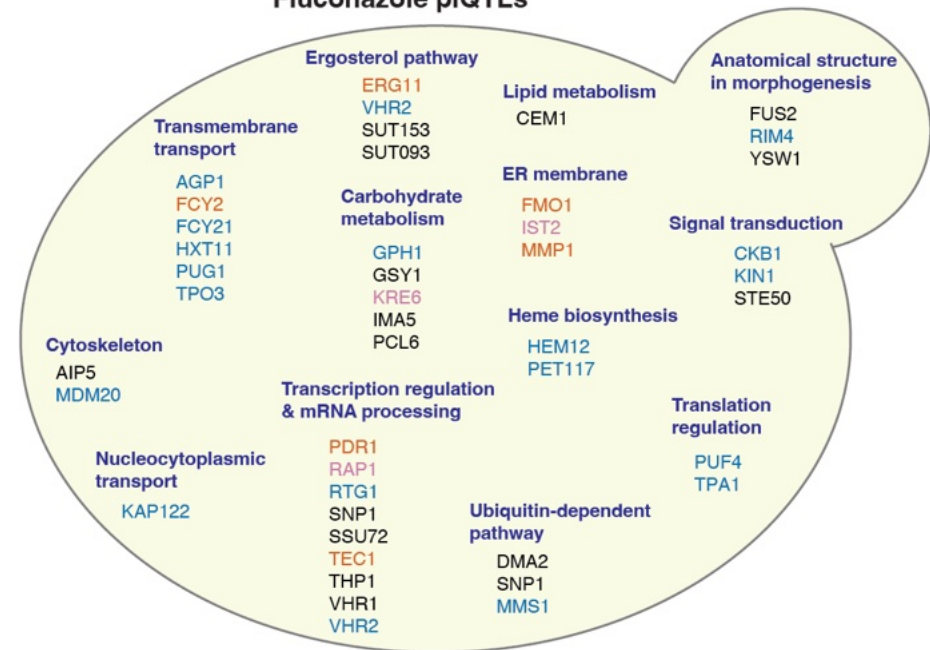


piQTLs capture known, and provide hypotheses for novel, drug mechanisms

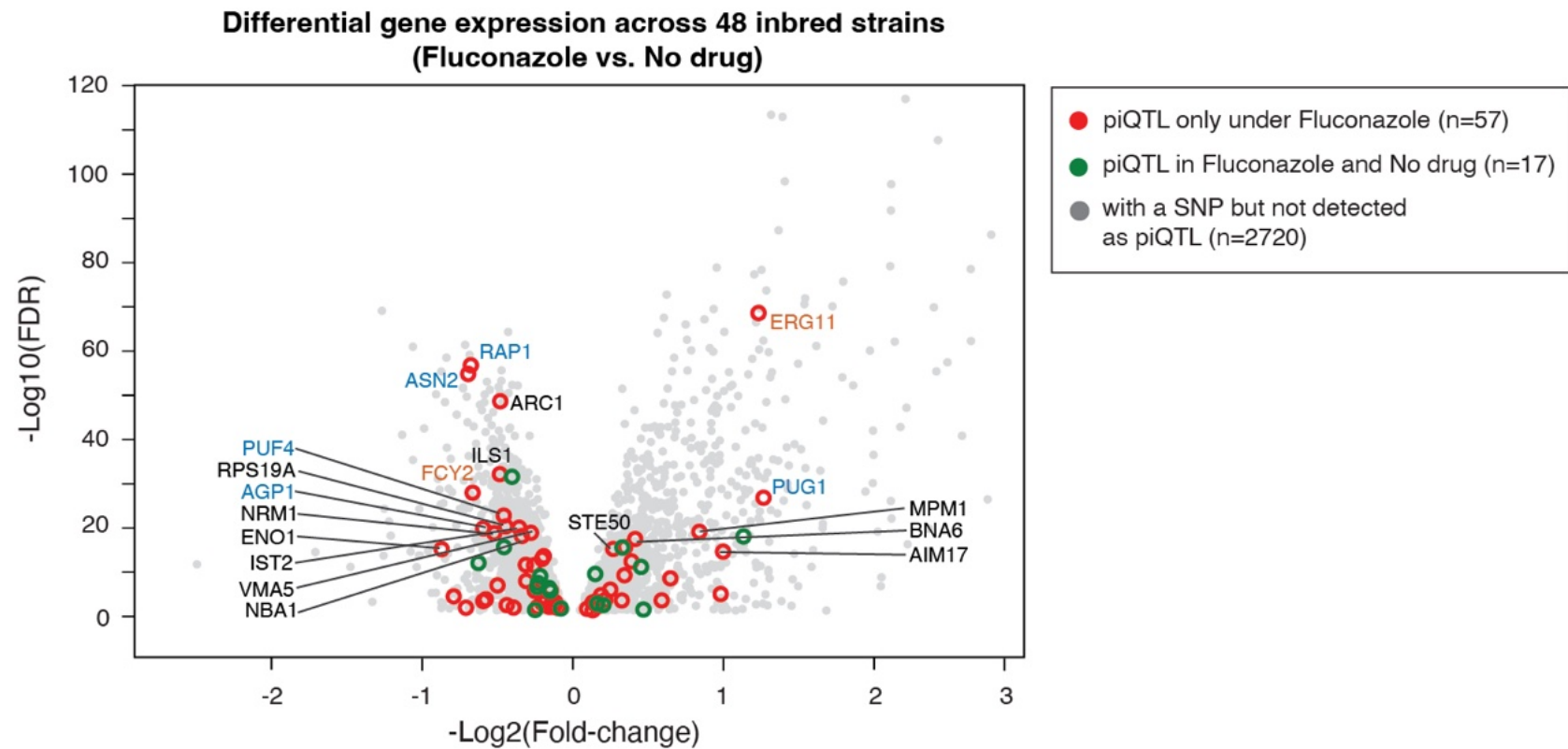
Co-citation of gene and drug in PubMed



Fluconazole piQTLs



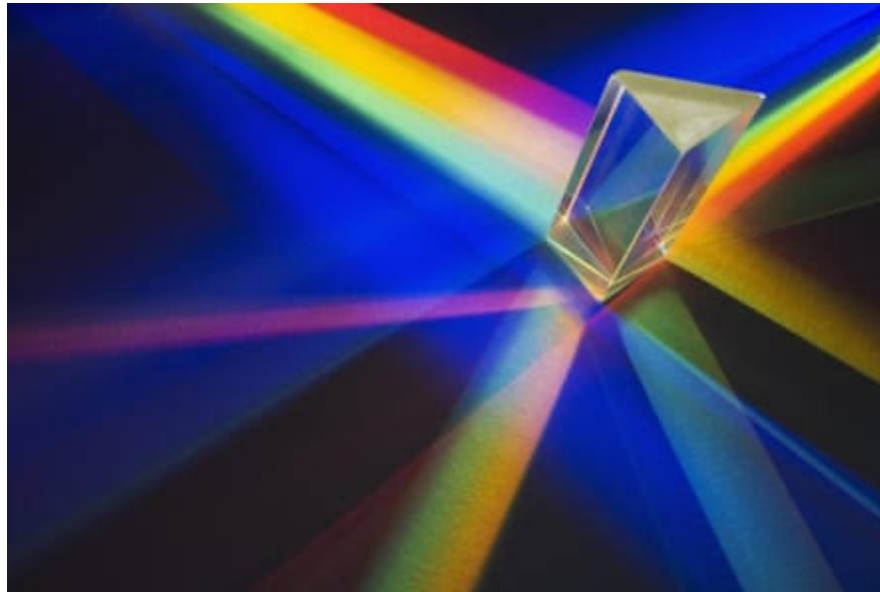
piQTLs capture known, and provide hypotheses for novel, drug mechanisms



Summary

- piQTL mapping is an accurate reporter of GxE
- Sensitivity of piQTL reflects the small world topology of PPI networks
- PPI is dominated by trans SNPs, revealing regulatory mechanisms
- PPI reveals functionality of non-coding regions including ncRNAs
- PPI could provide quantitative measurements of pleiotropy and epistasis of complex phenotypes

Pooled Recombinant Integration of Seamless Markers

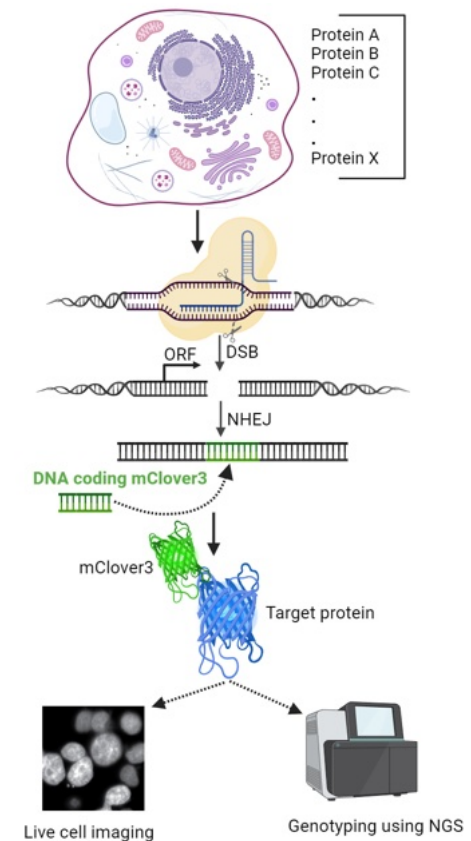
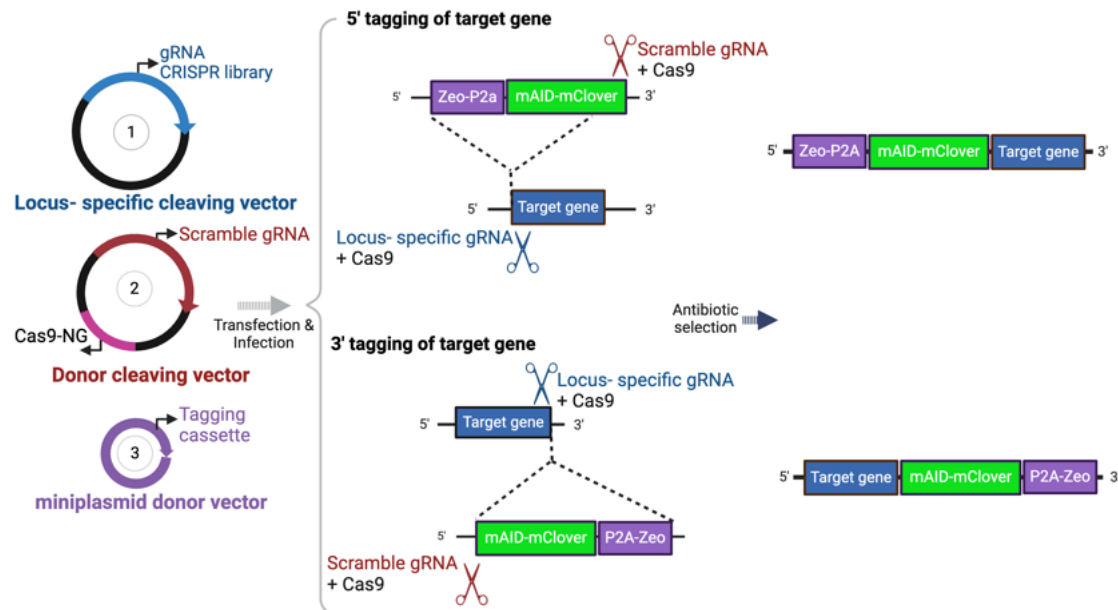


Minoo Karimi

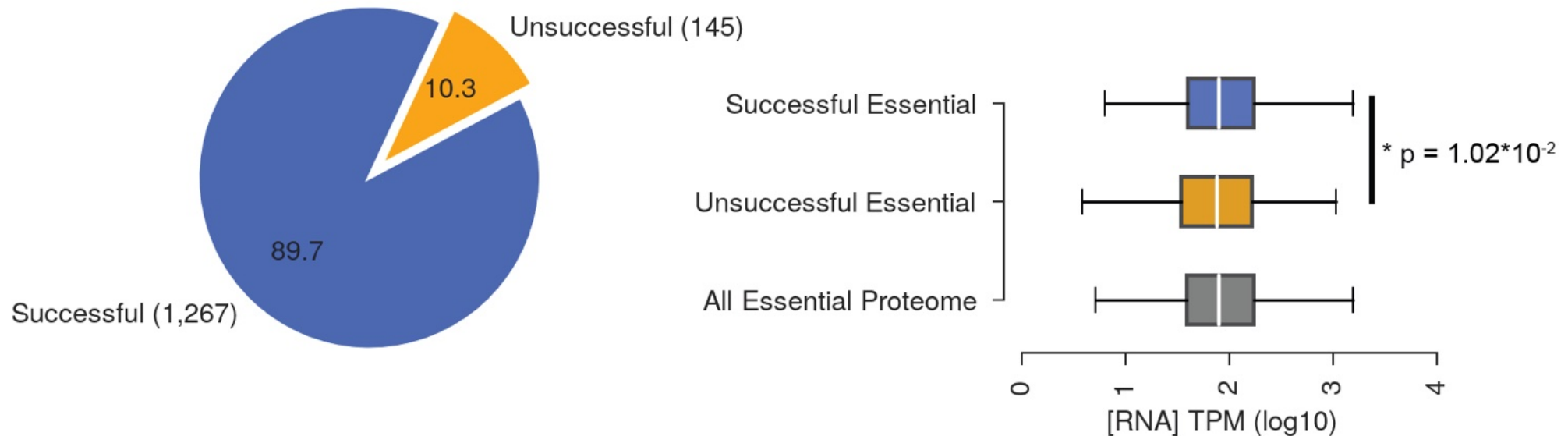


Bringing the full spectrum of proteome manipulation and measurement to everyone

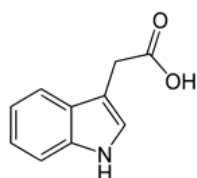
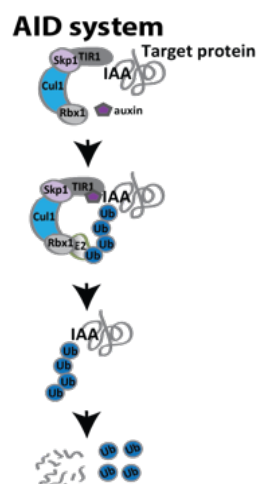
PRISM to integrate tags into yeast or human cells



NGS results indicate that ~90% of the essential genes were tagged 5' or 3' to protein-coding sequences using PRISM

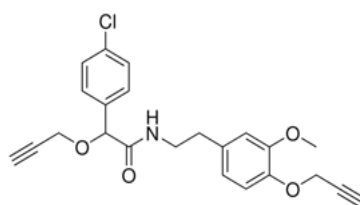
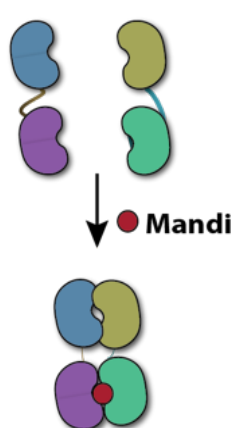


Tags to manipulate or measure protein and protein-protein interaction abundances



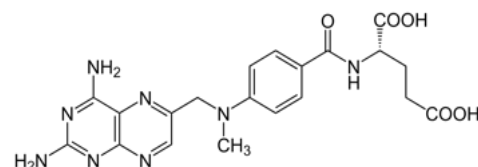
Yesbolatova, et al., *Nat. Comm.*, 2020

Mandi-CIP



Ziegler, et al., *Nat. Chem. Biol.*, 2022

DHFR PCA



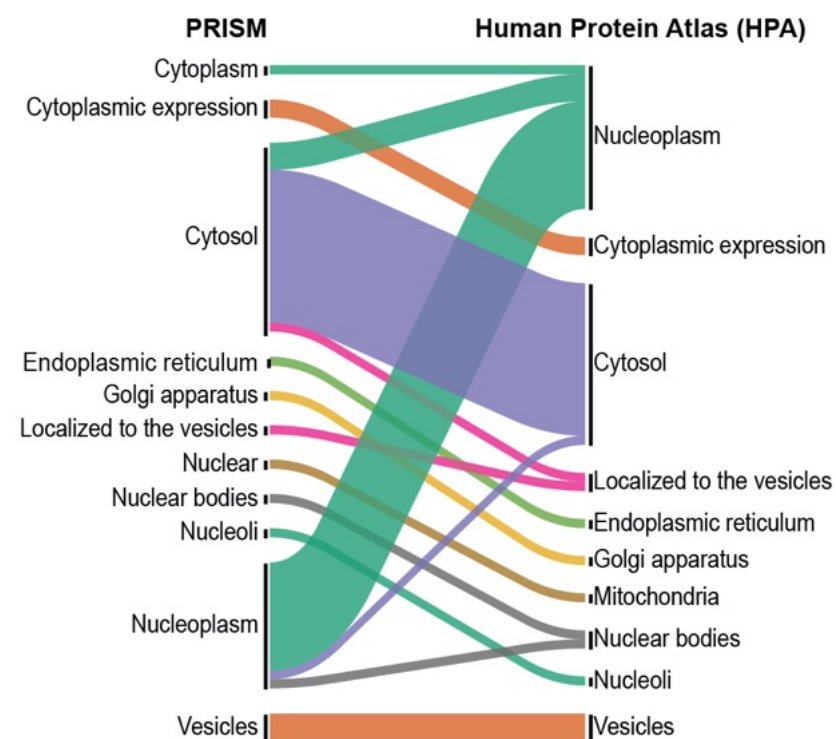
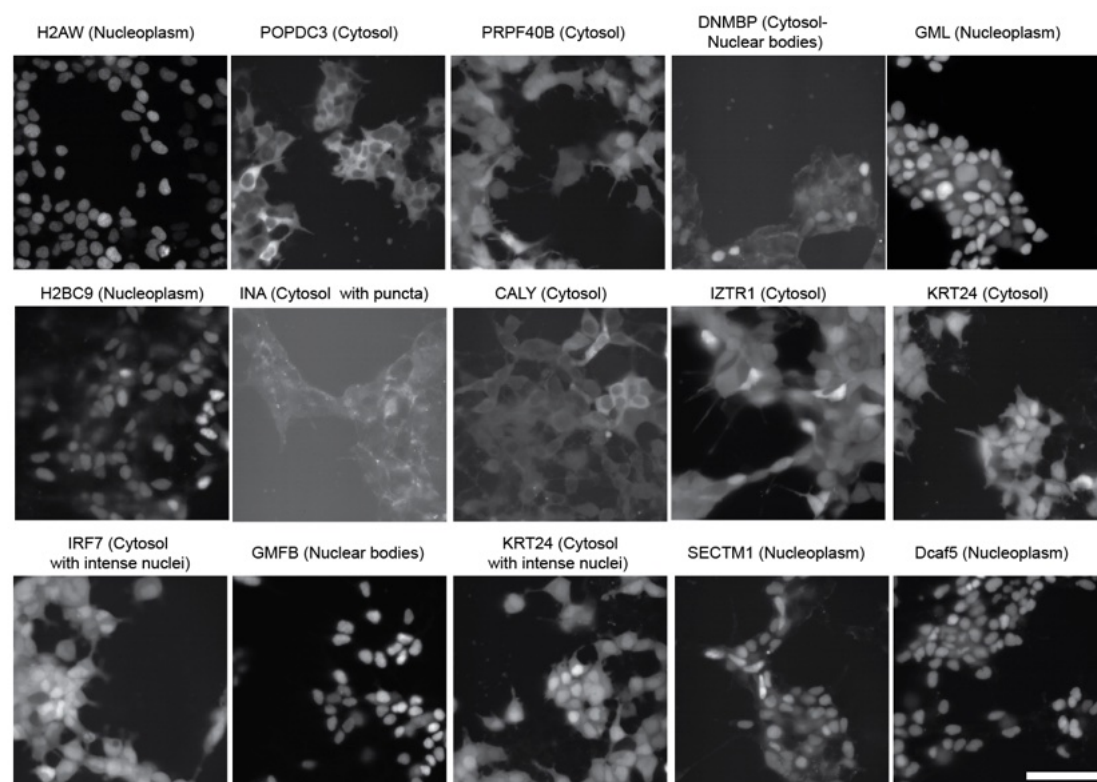
Tarassov, et al., *Science* 2008

Abund. DHFR PCA



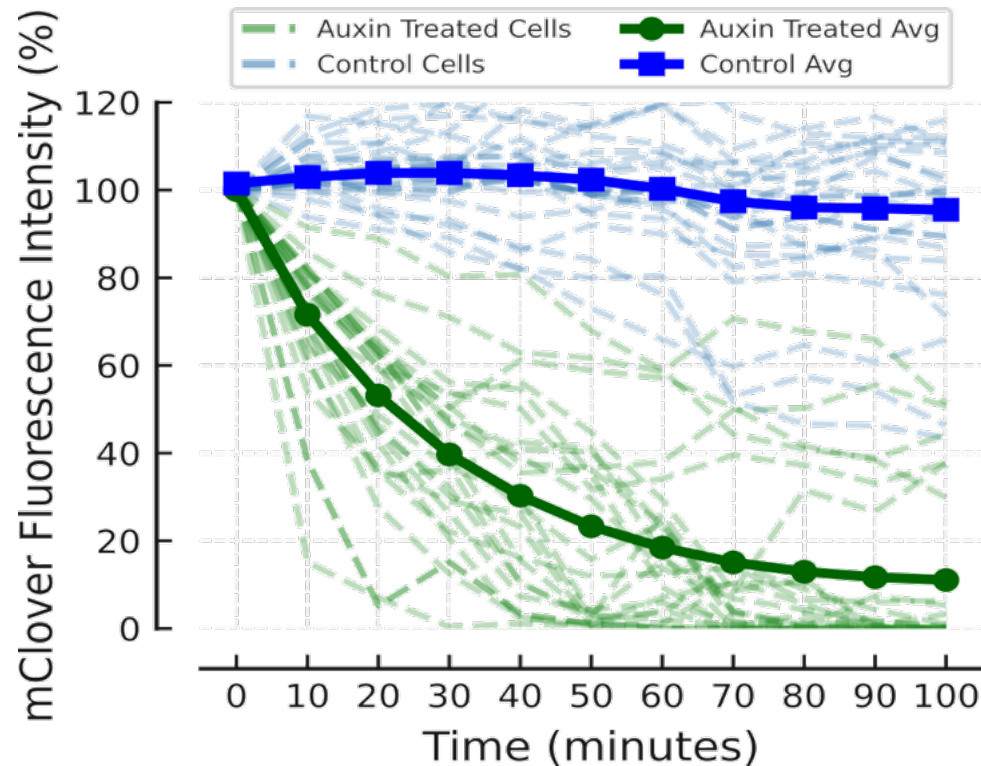
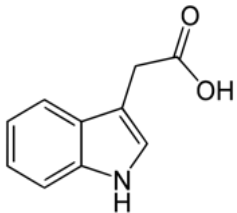
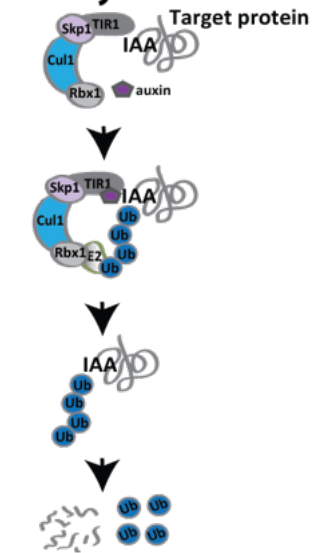
Levy, et al., *Cell Rep.*, 2015

Localizing of Proteins labelled with GFP by PRISM is consistent with immunofluorescence localization



Auxin-mediated large scale protein degradation performed with PRISM

AID system



What's next?

- piQTL of human Yoruba lymphoblastoid cell lines
- Seeking collaborations for other human piQTL projects
- Rewiring of disease associated PPI networks during differentiation of iPS
- Induced rewiring of PPIs by Proteome-Scale Chemical-Induced Proximity (ProSCIP) for cancer drug discovery
- Proteome-wide conditional gene knockdown screens
- Luciferase-based PCA for single cell detection of PPI by luminescence-NGS readout.

Acknowledgements

Tatsuya Sakaguchi



Xavier Castellanos-Girouard



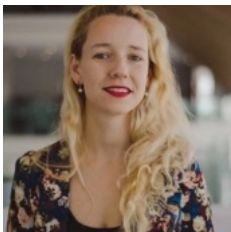
Savandara Besse



Minoo Karimi



Julie Hussin (Montreal Heart Institute)



Quantitative Cell Biology and Evolution SuperGroup (w/ Serohijos Lab)



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